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Bahreinian et al.

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(54) **OVEN**

(56)

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Robert Vandenham, Woodbridge (CA)

(73) Assignee: **NewAge Products Inc.**, Toronto (CA)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 1030 days.

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Attorneys PLLC

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ABSTRACT

An oven includes upper and lower assemblies, a fire box and a heat absorbing and emanating device. The upper assembly forms a cooking chamber and has a front side with an open entry to the cooking chamber. The lower assembly is located below the upper assembly and has an internal cavity. The fire box is located within the internal cavity and includes a heat source. The device is located between the upper and lower assemblies and absorbs heat from the heat source and emanates heat into the cooking chamber of the upper assembly. The upper assembly includes inner and outer dome assemblies. The inner dome is located within the outer dome and forms first and second airflow pathways.

22 Claims, 36 Drawing Sheets

(65) **Prior Publication Data**

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(51) **Int. Cl.**

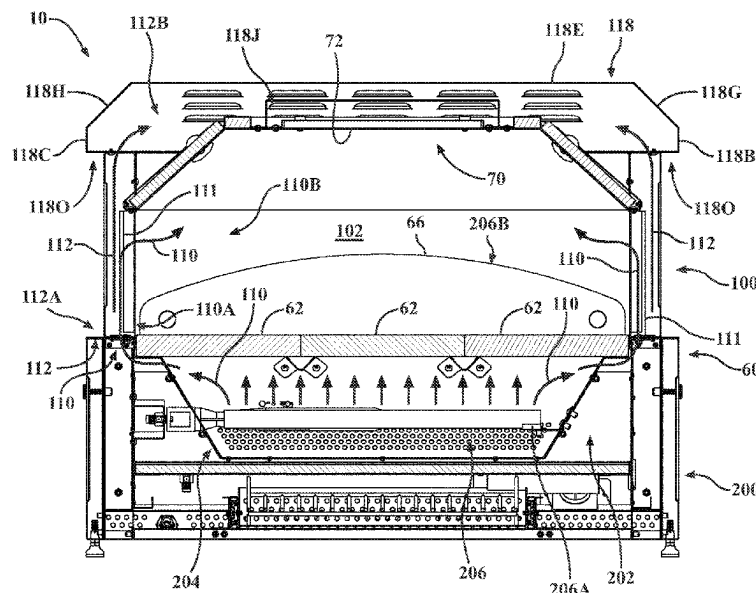
F24C 15/32 (2006.01)
A21B 1/10 (2006.01)
A21B 1/26 (2006.01)
F24C 15/02 (2006.01)
F24C 15/24 (2006.01)
F24C 15/34 (2006.01)
F24C 15/36 (2006.01)

(52) **U.S. Cl.**

CPC **A21B 1/10** (2013.01); **A21B 1/26**
(2013.01); **F24C 15/24** (2013.01); **F24C 15/32**
(2013.01); **F24C 15/34** (2013.01); **F24C 15/36**
(2013.01); **F24C 15/02** (2013.01)

(58) **Field of Classification Search**

USPC D7/348; 126/273.5
See application file for complete search history.



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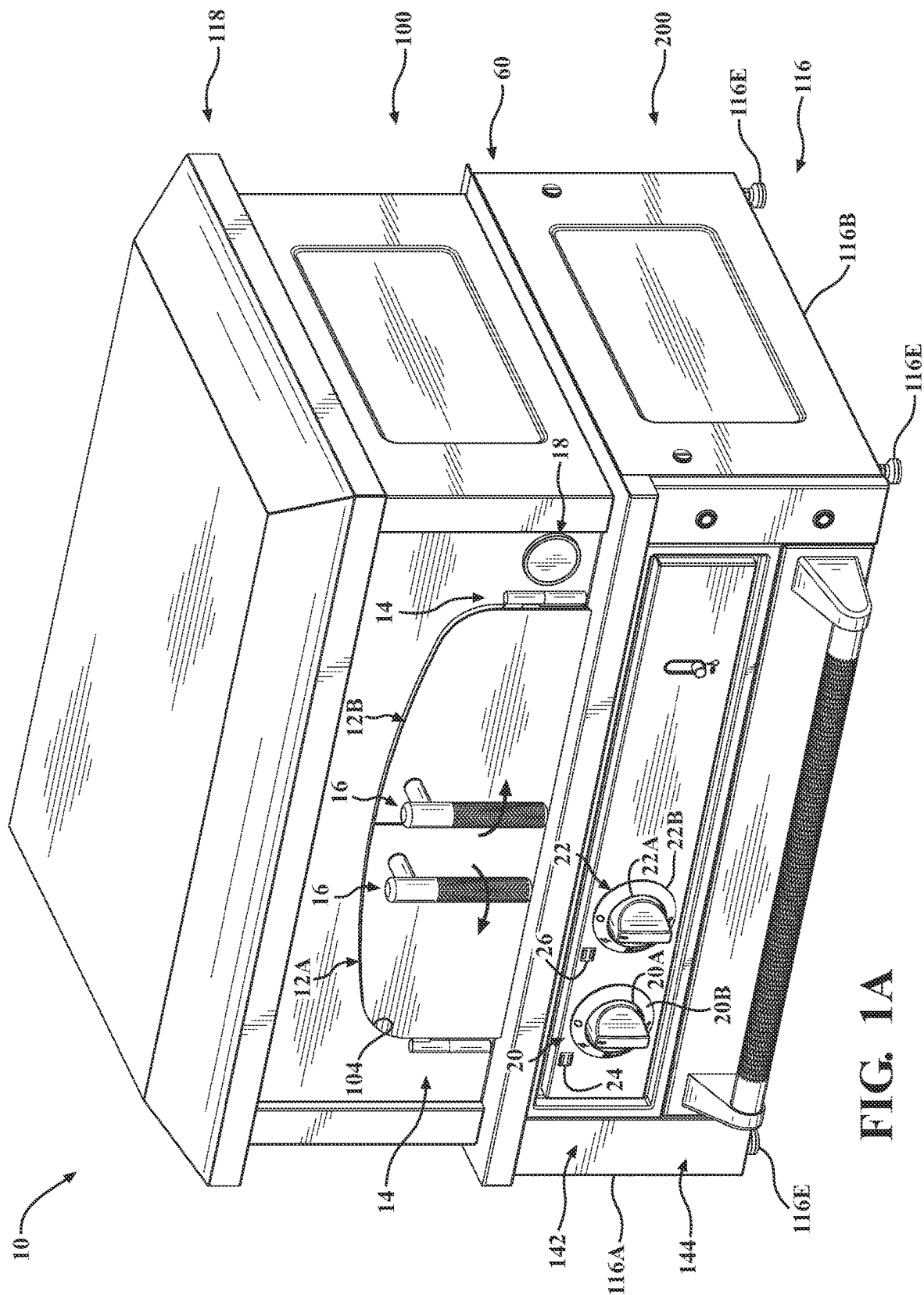


FIG. 1A

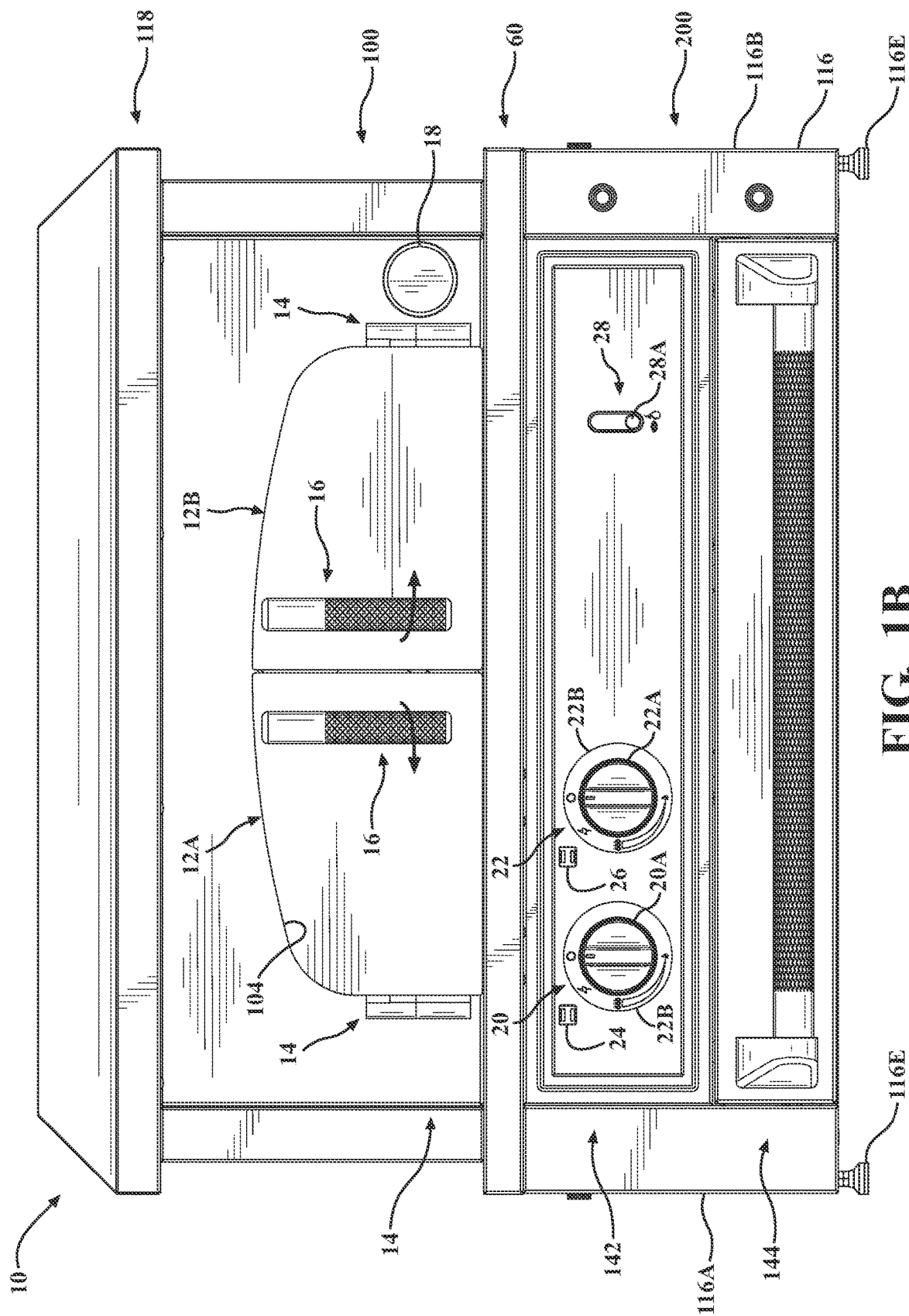


FIG. 1B

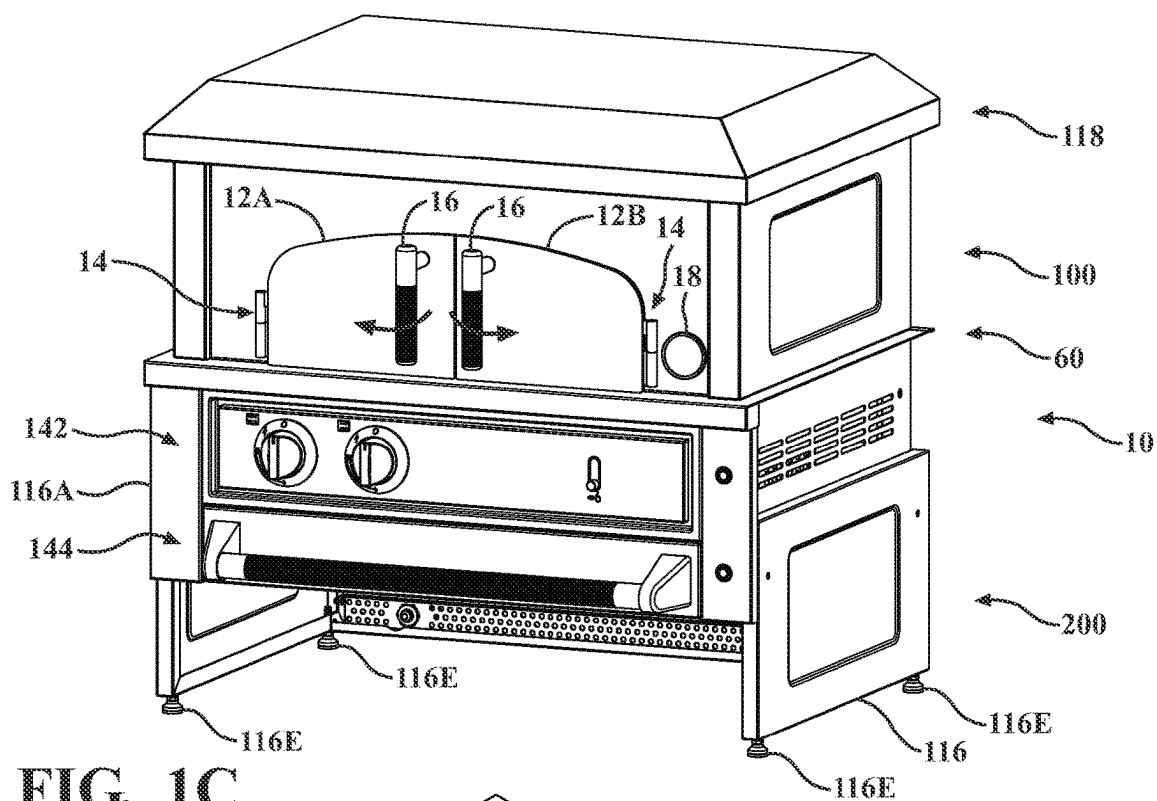


FIG. 1C

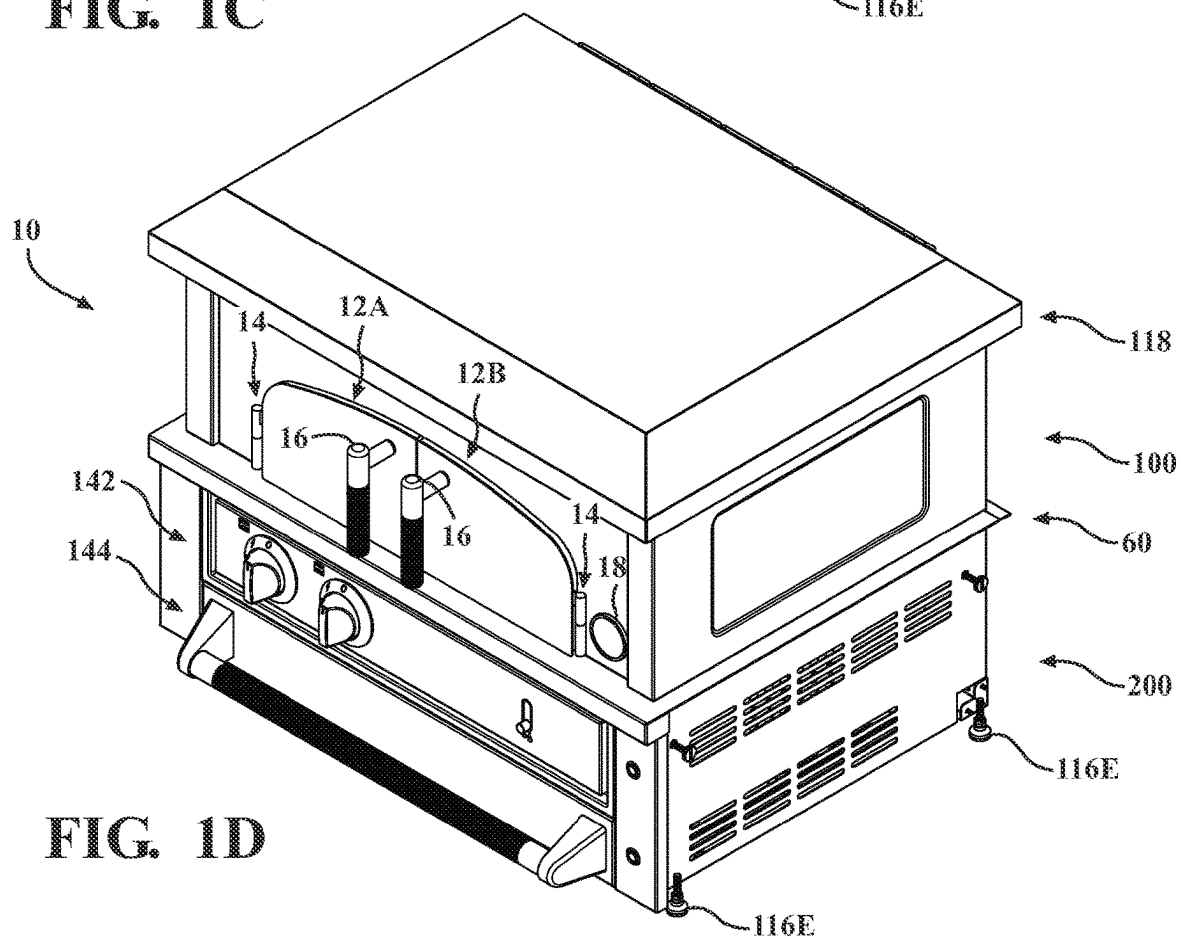
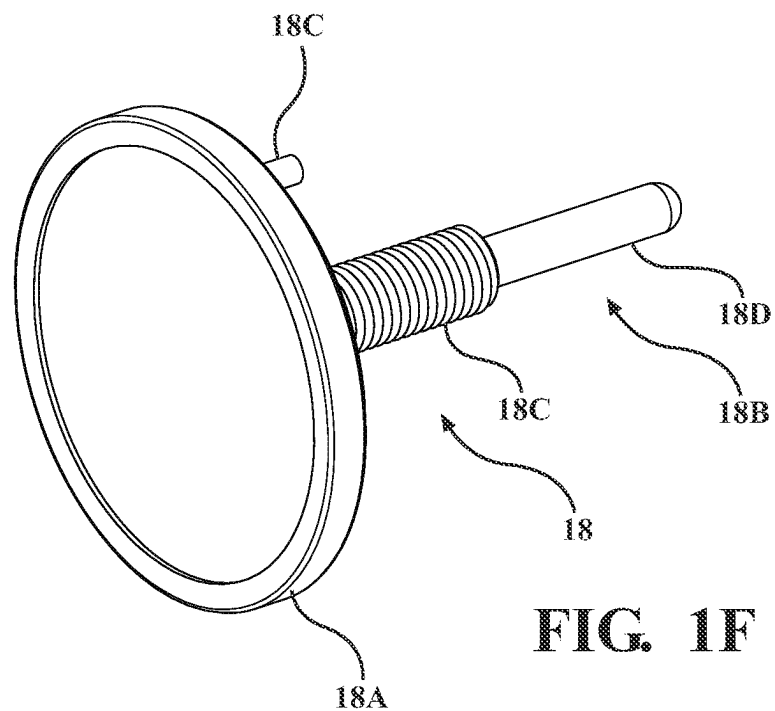
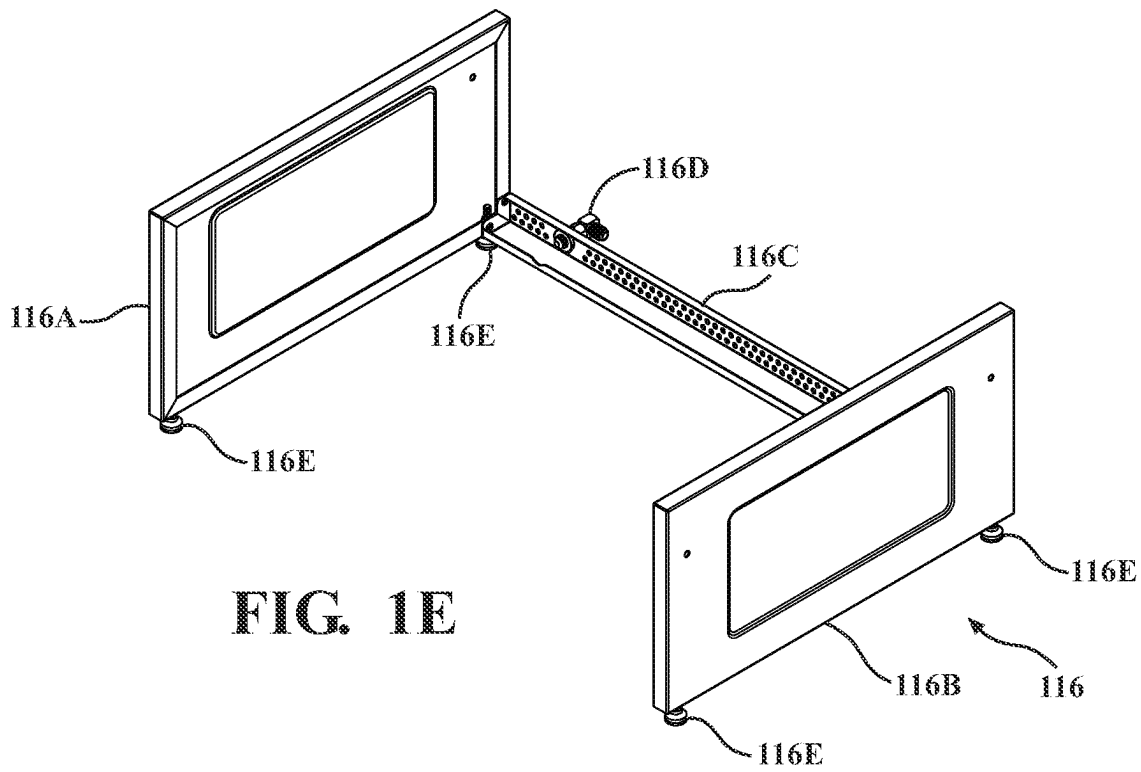


FIG. 1D



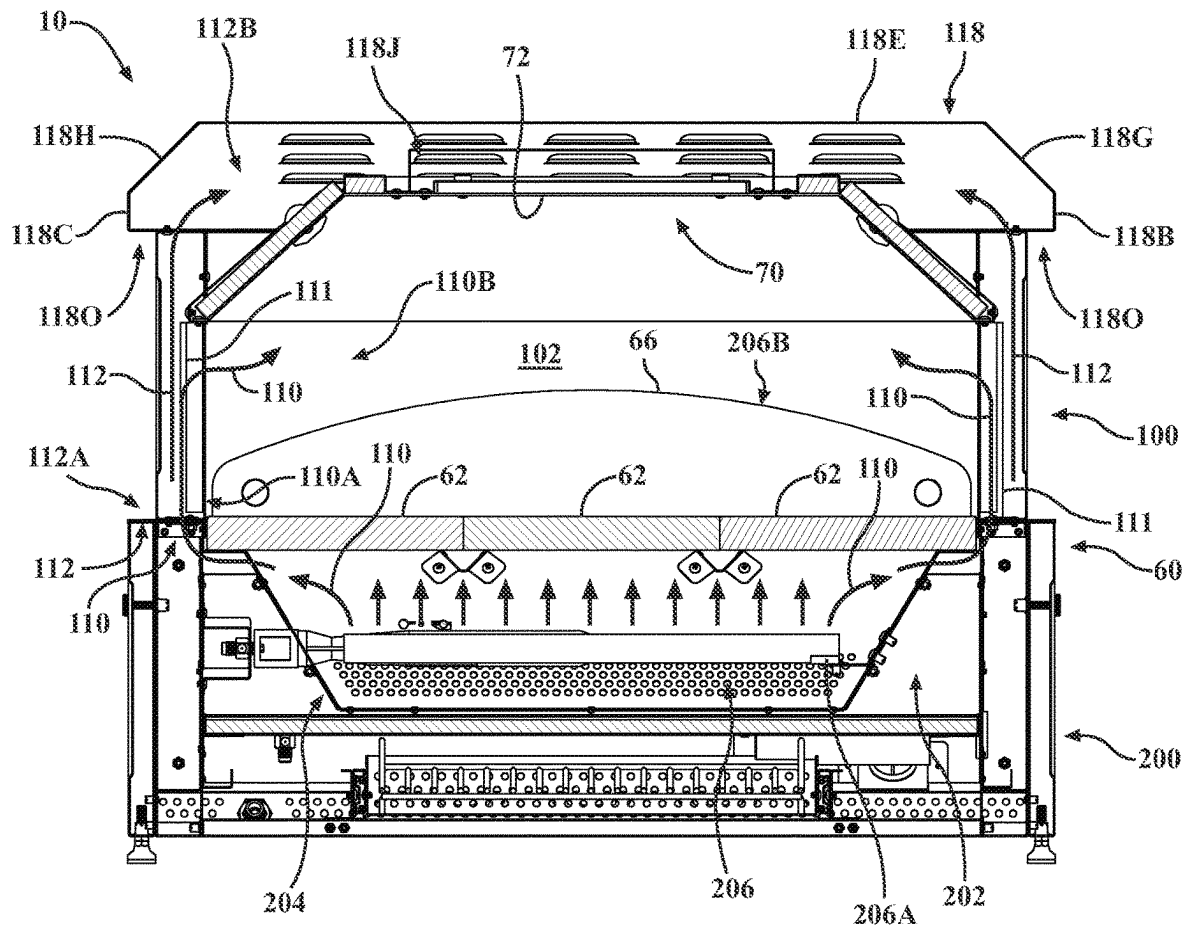


FIG. 2A

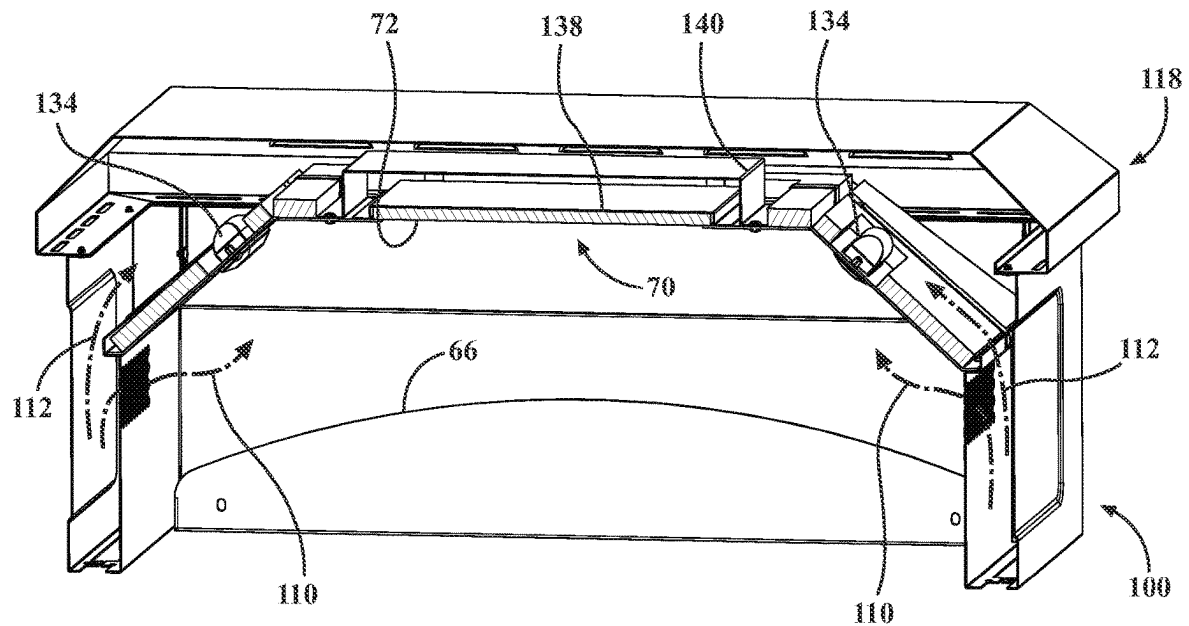


FIG. 2B

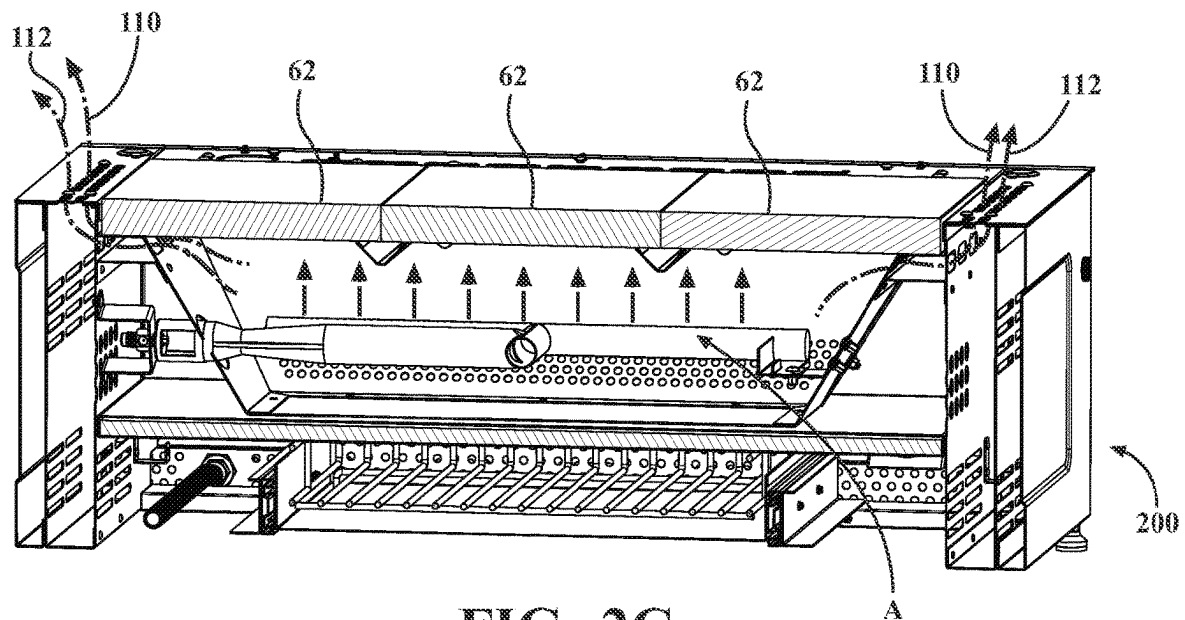
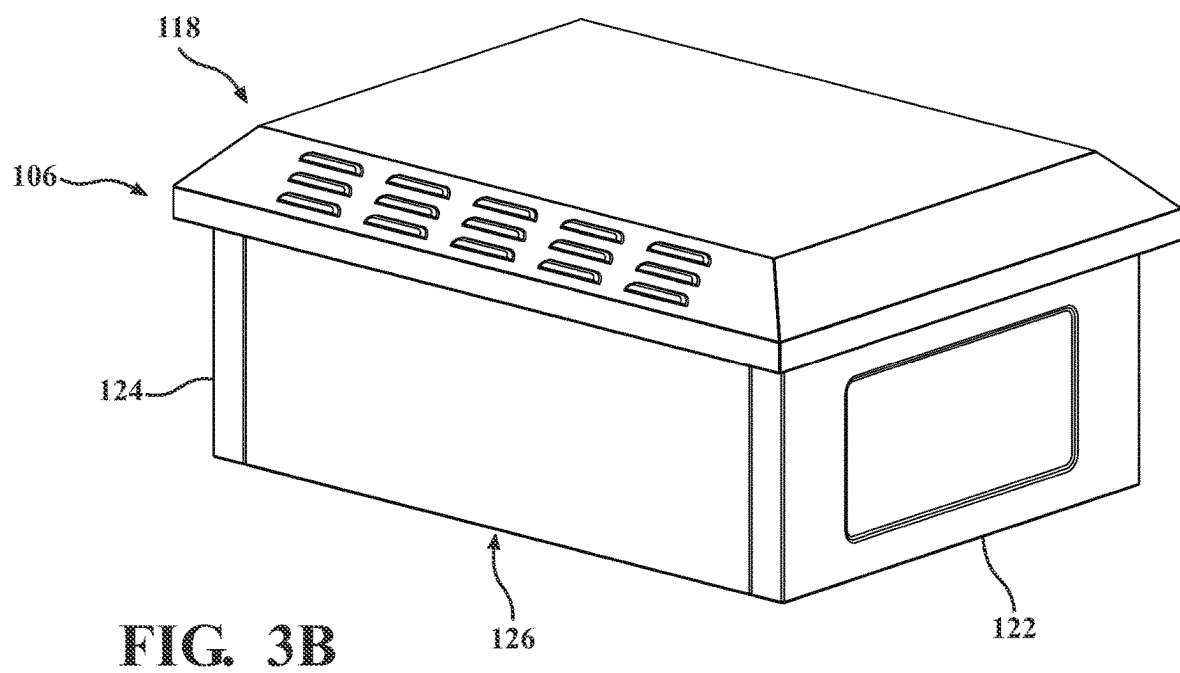
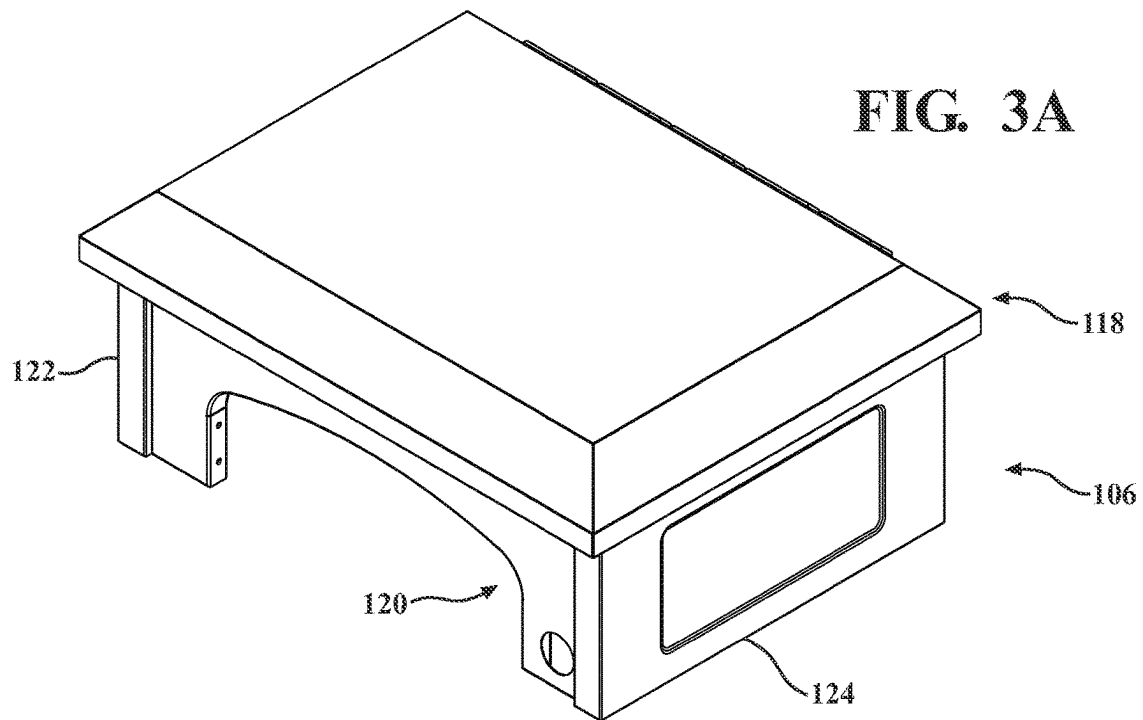


FIG. 2C



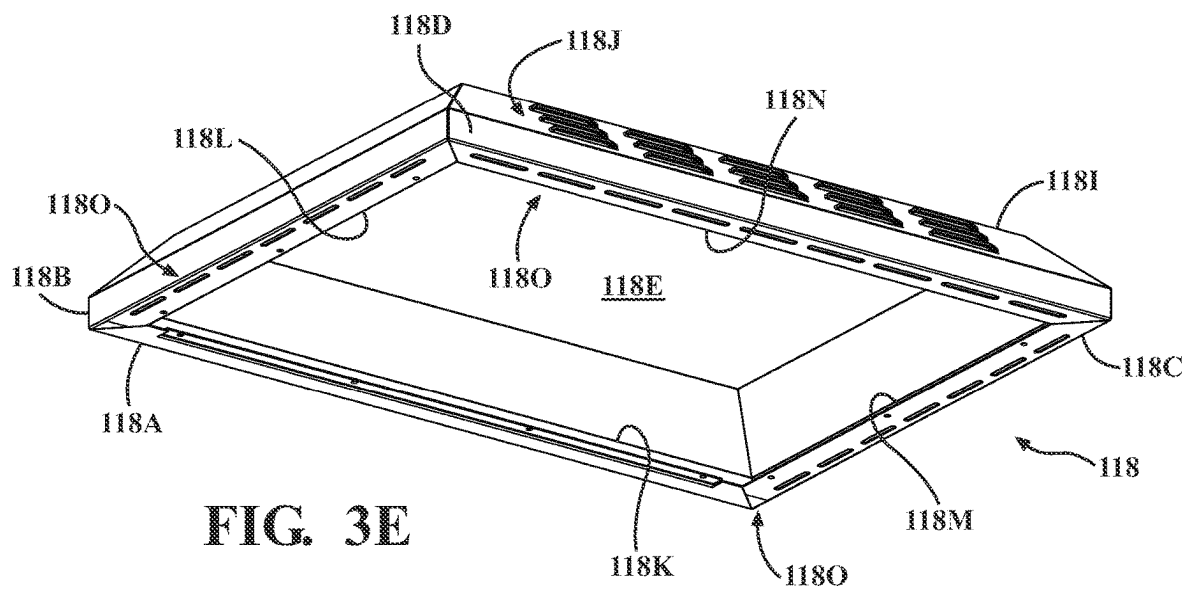
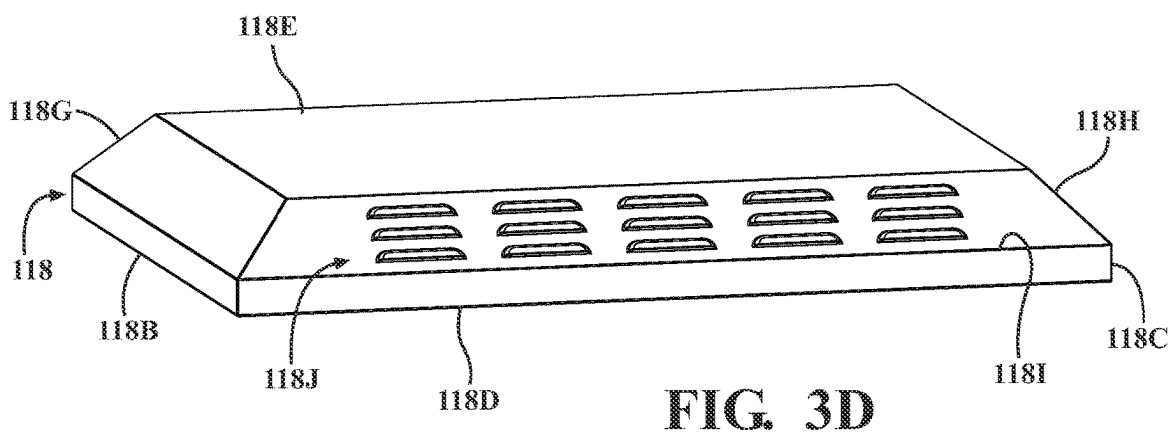
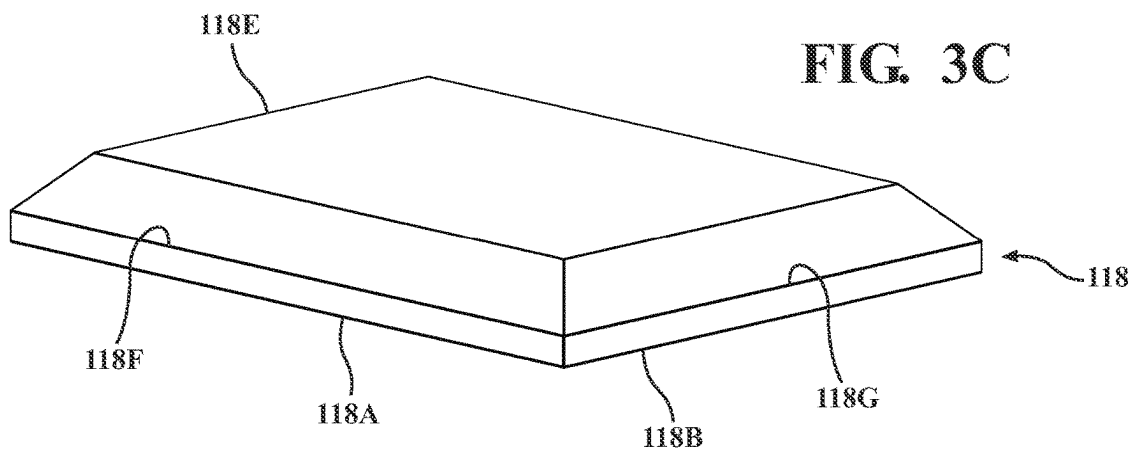


FIG. 3F

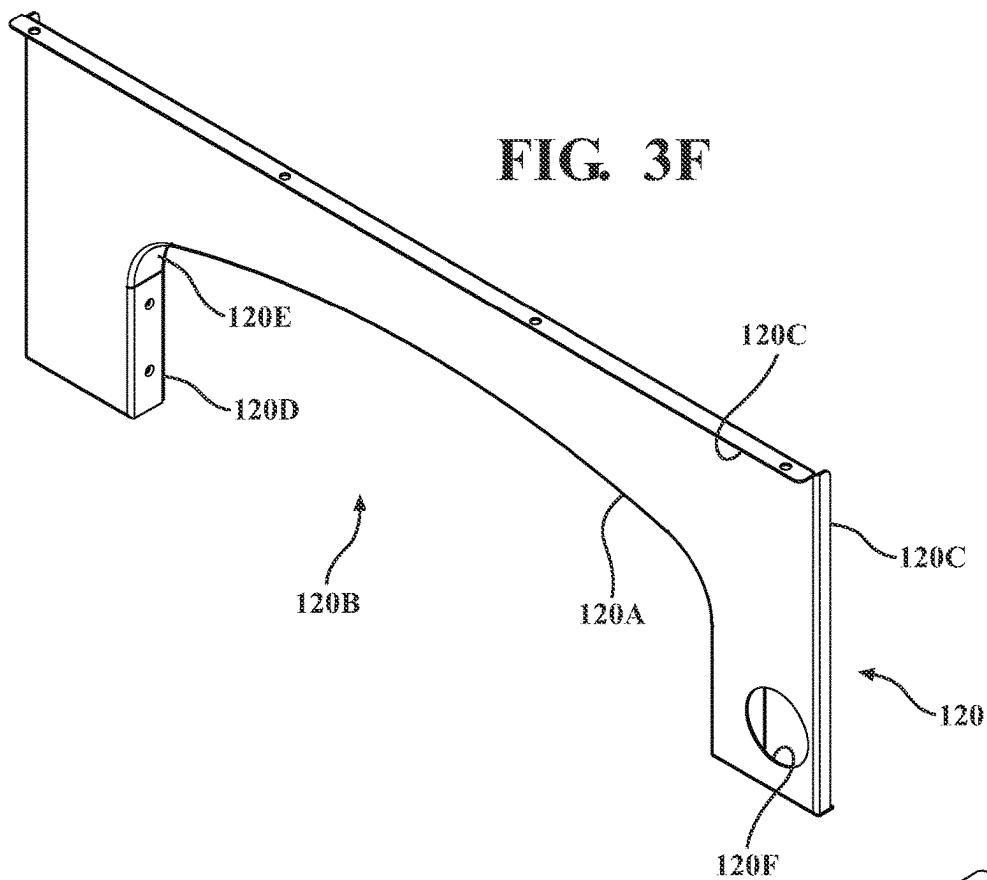
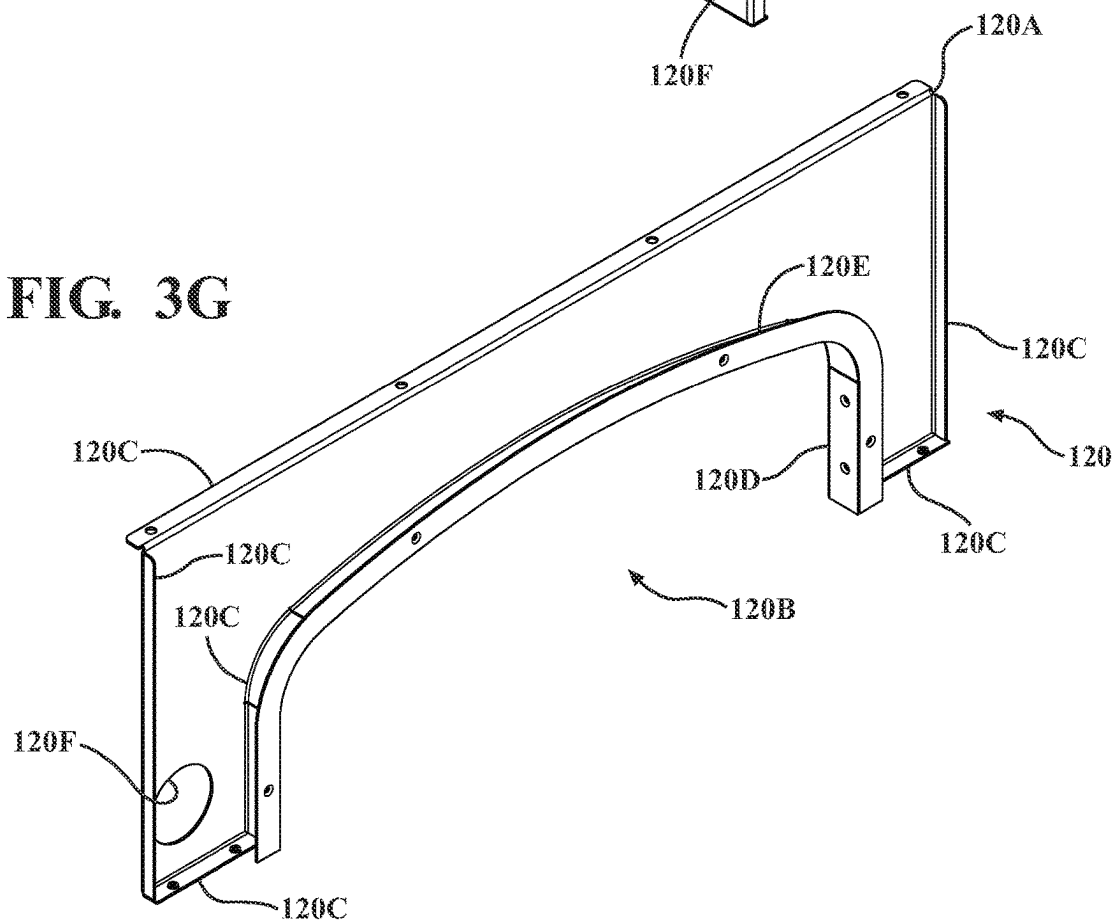
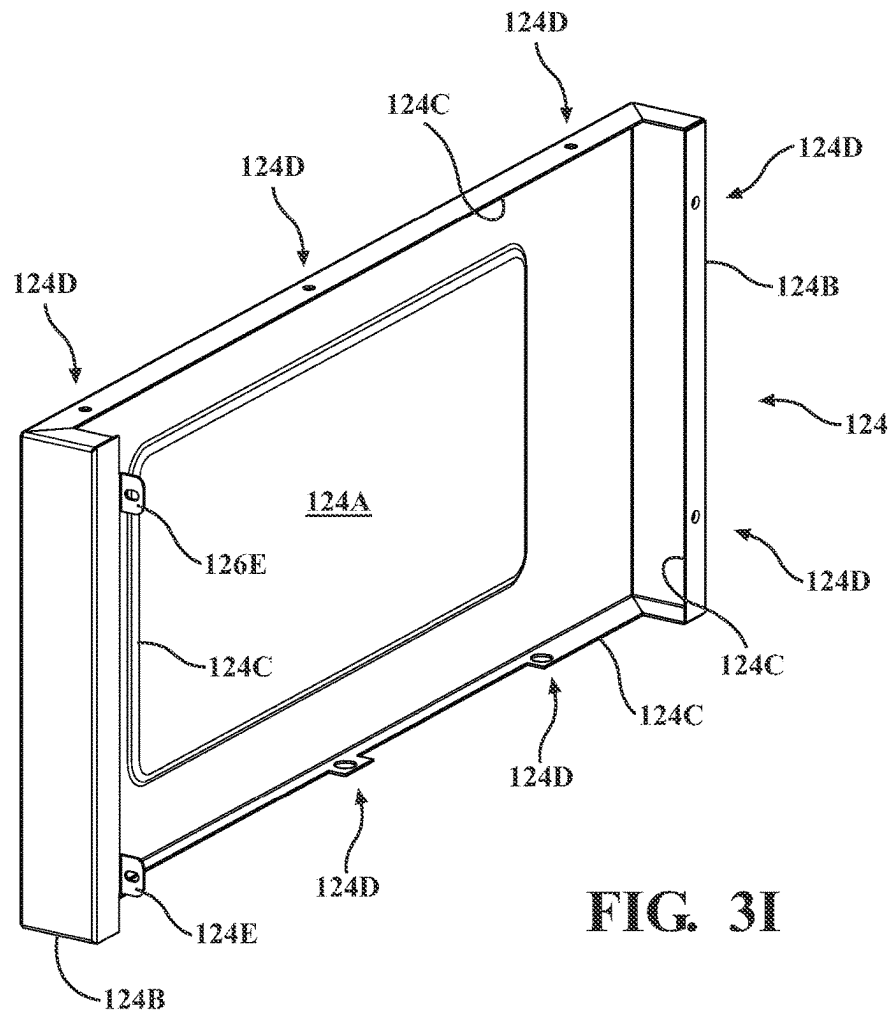
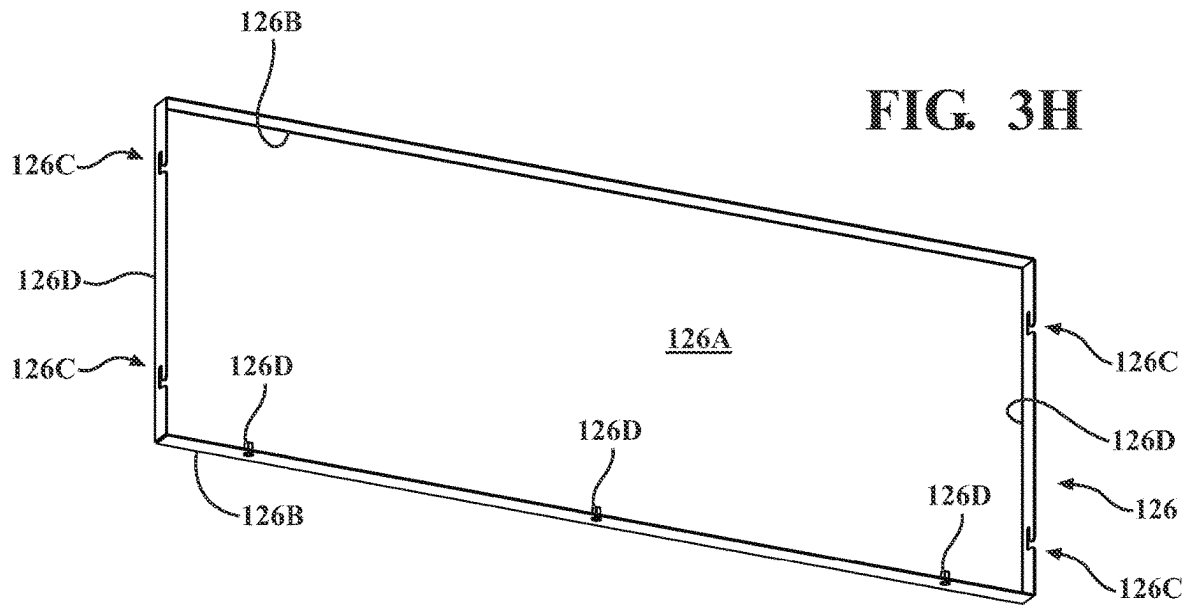
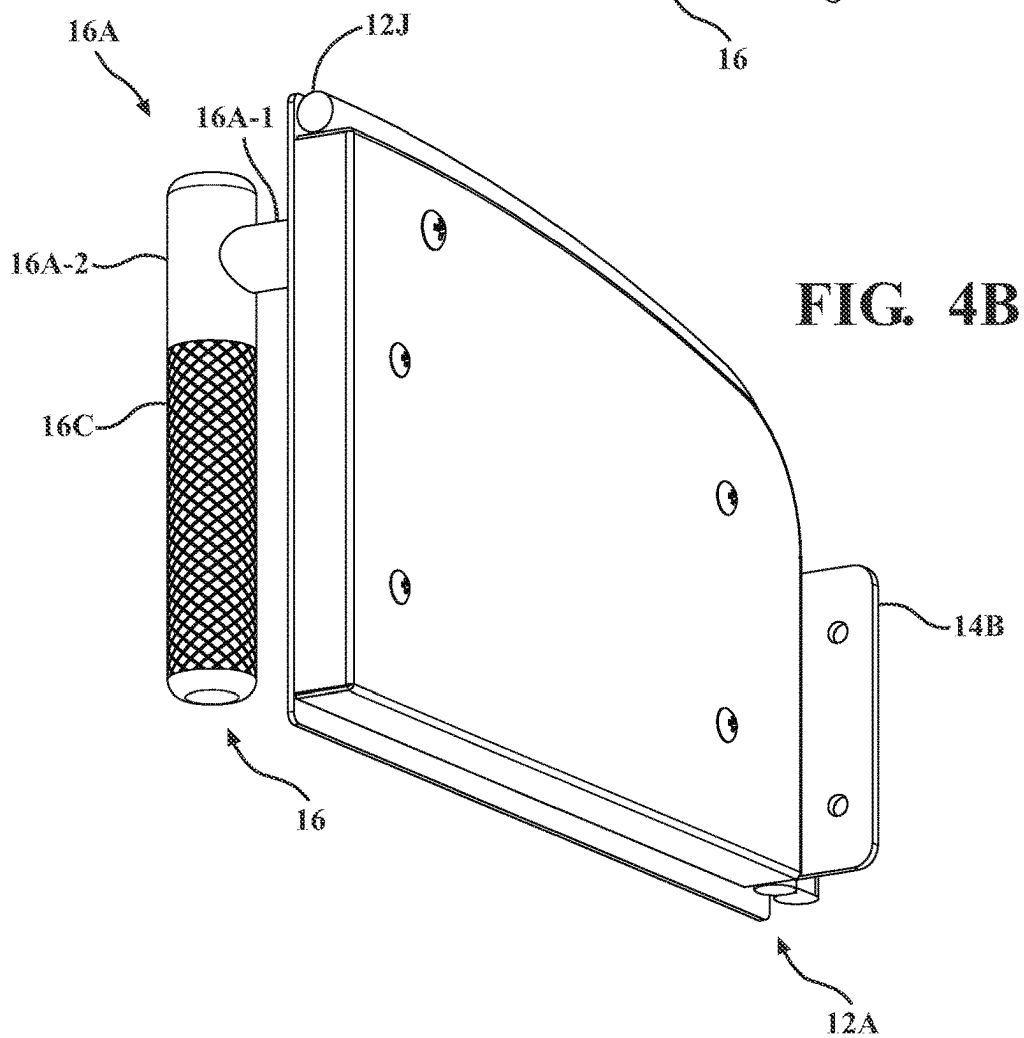
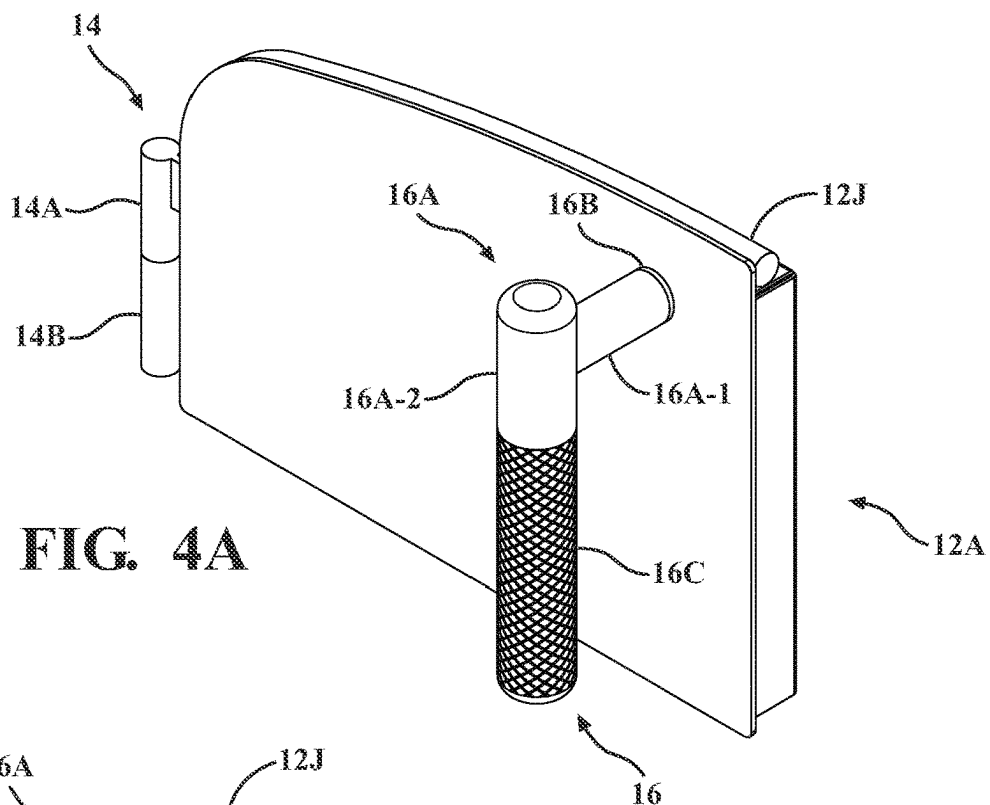
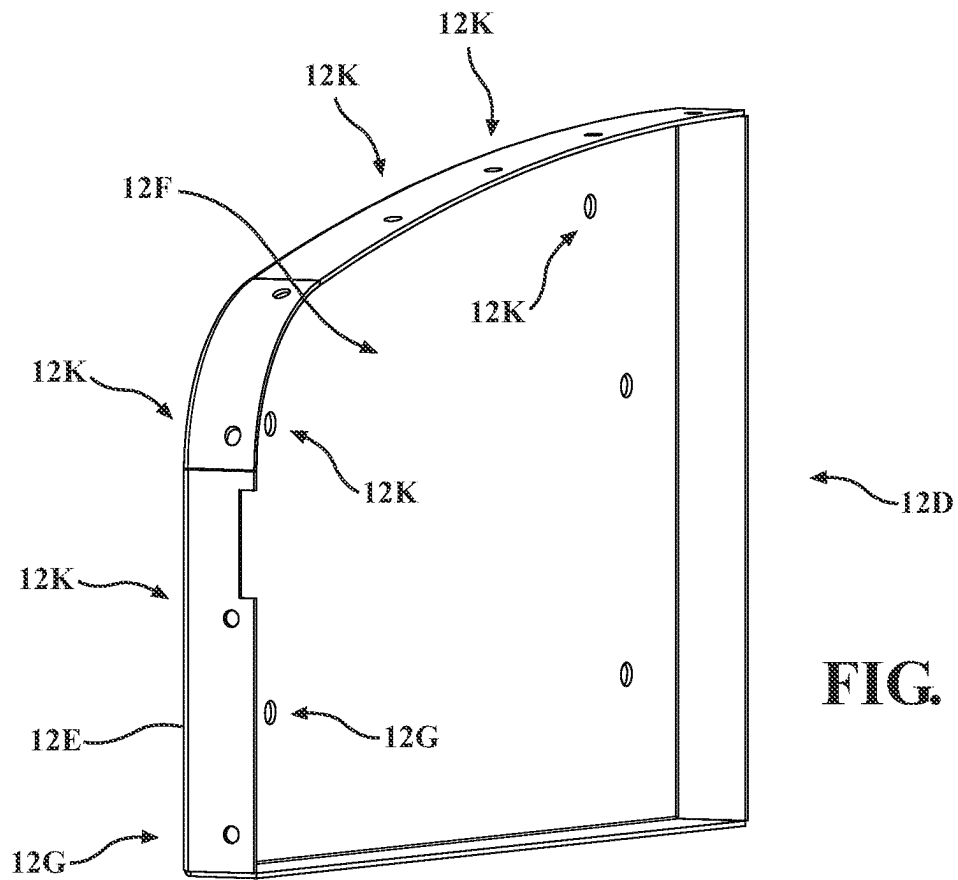
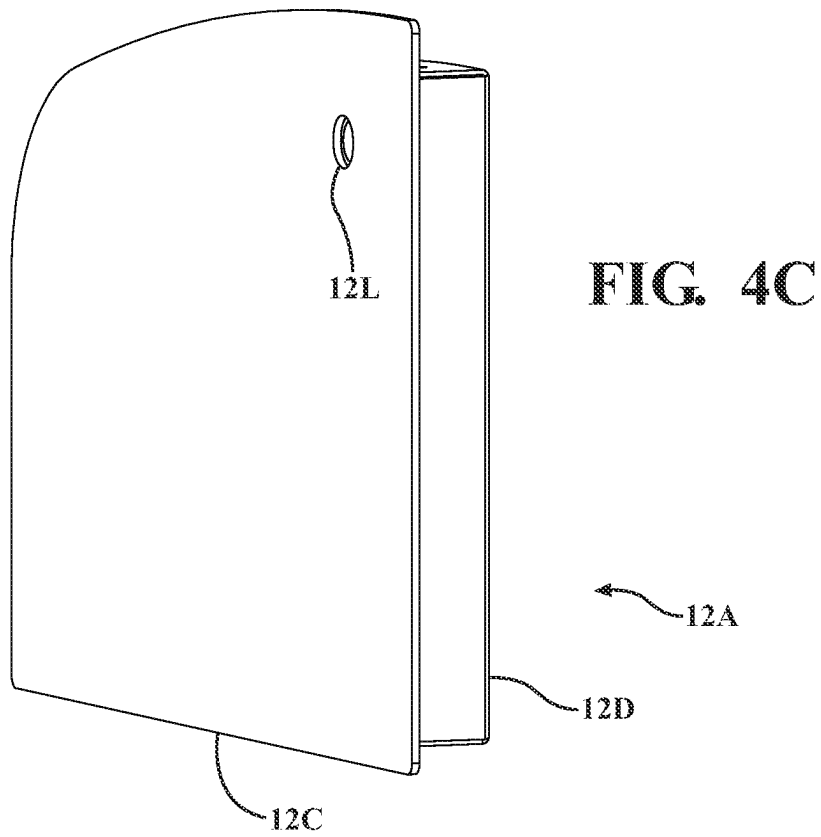


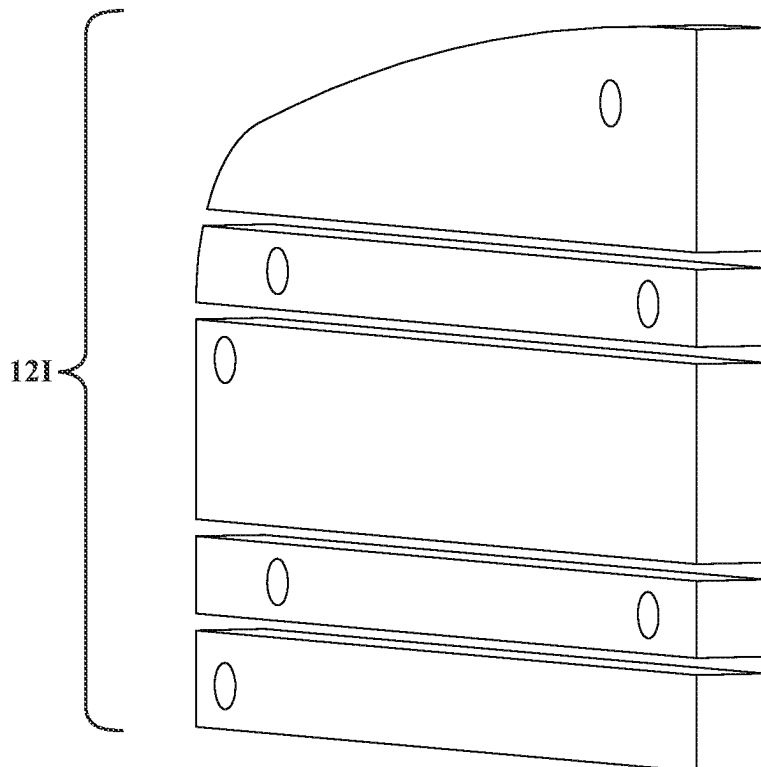
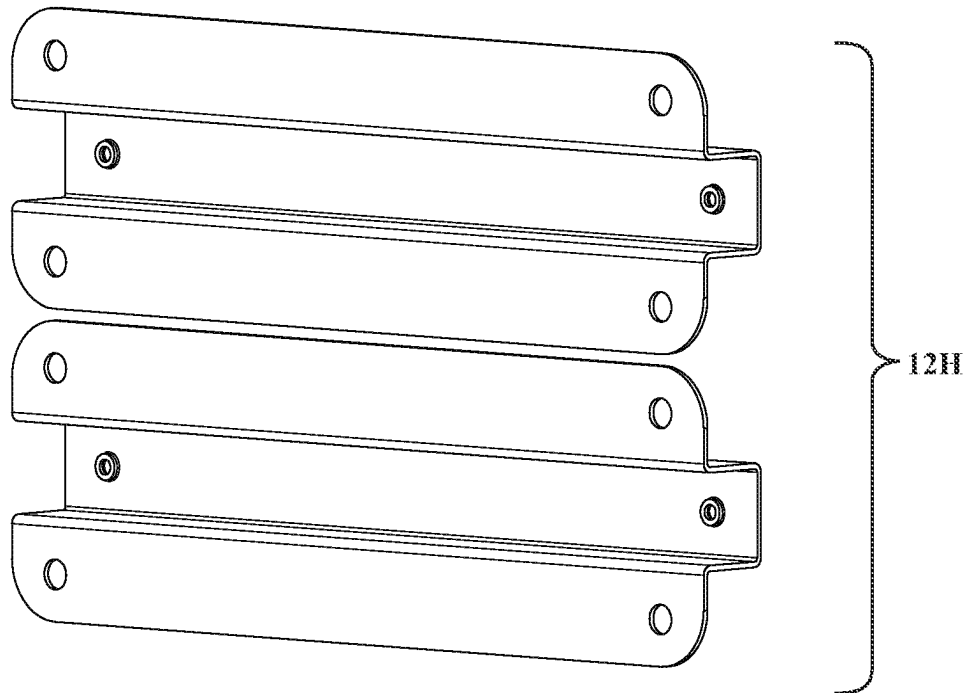
FIG. 3G











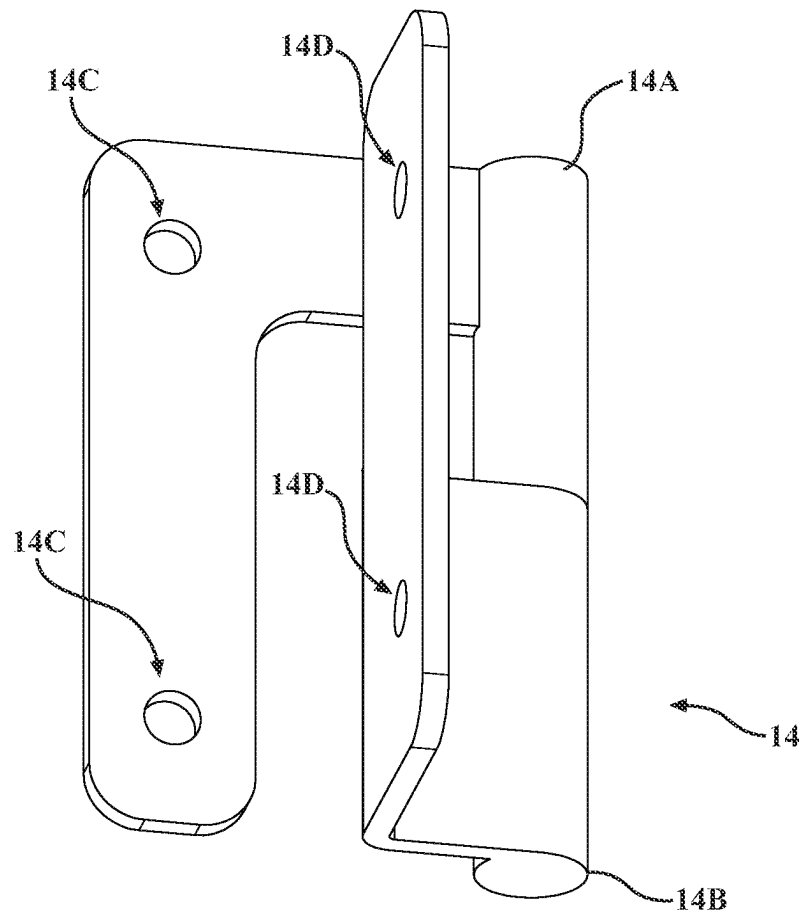
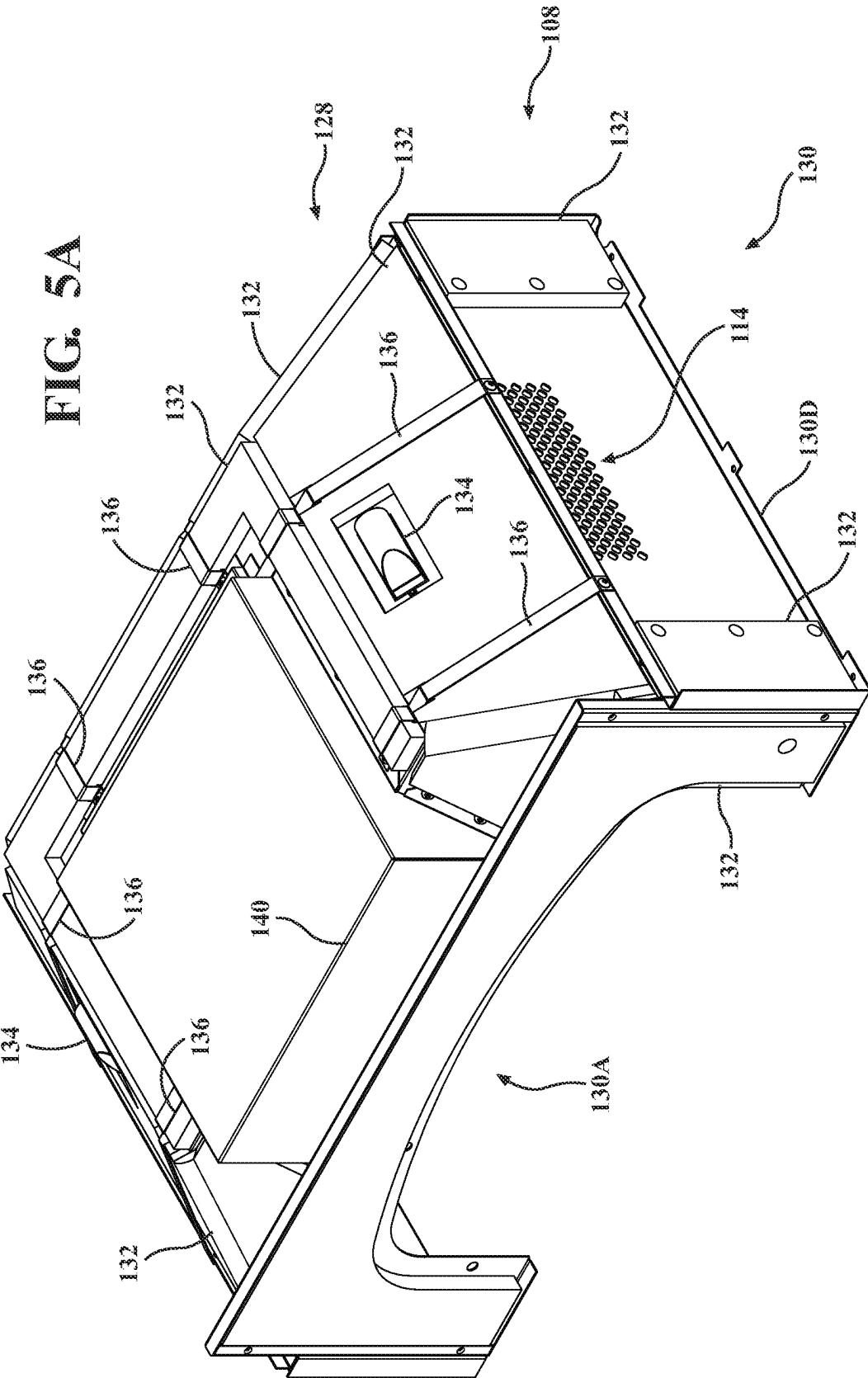


FIG. 4G



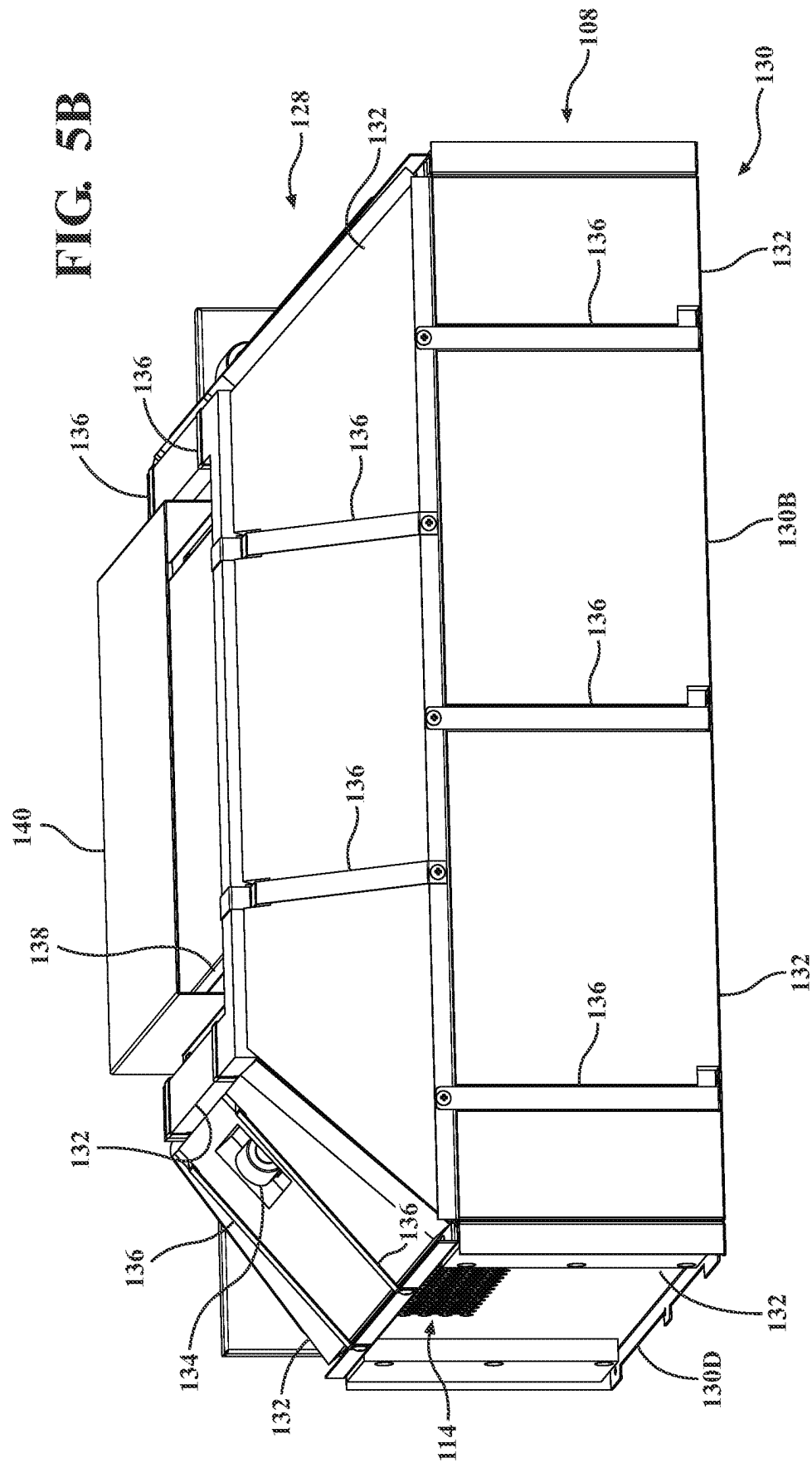
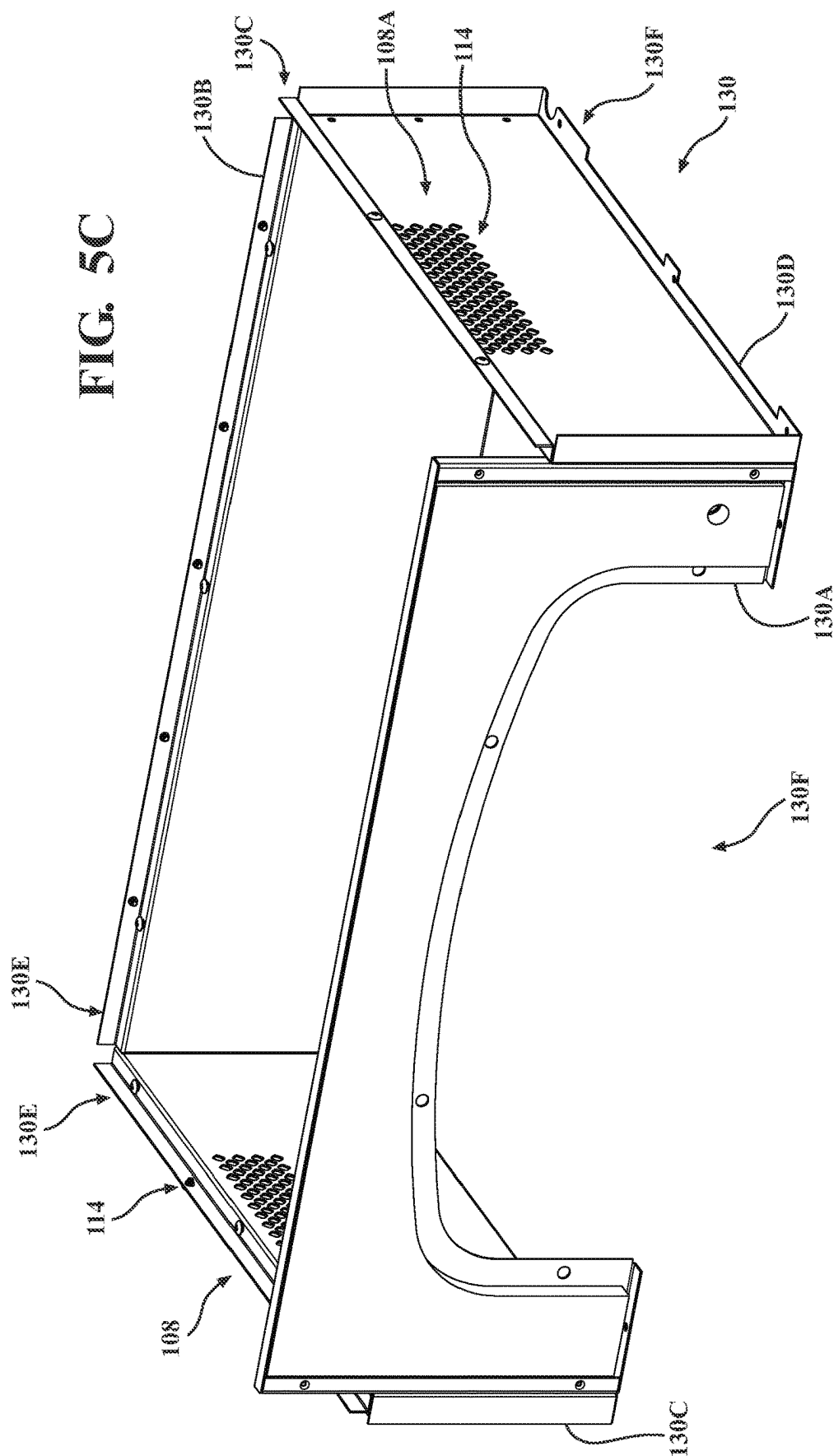
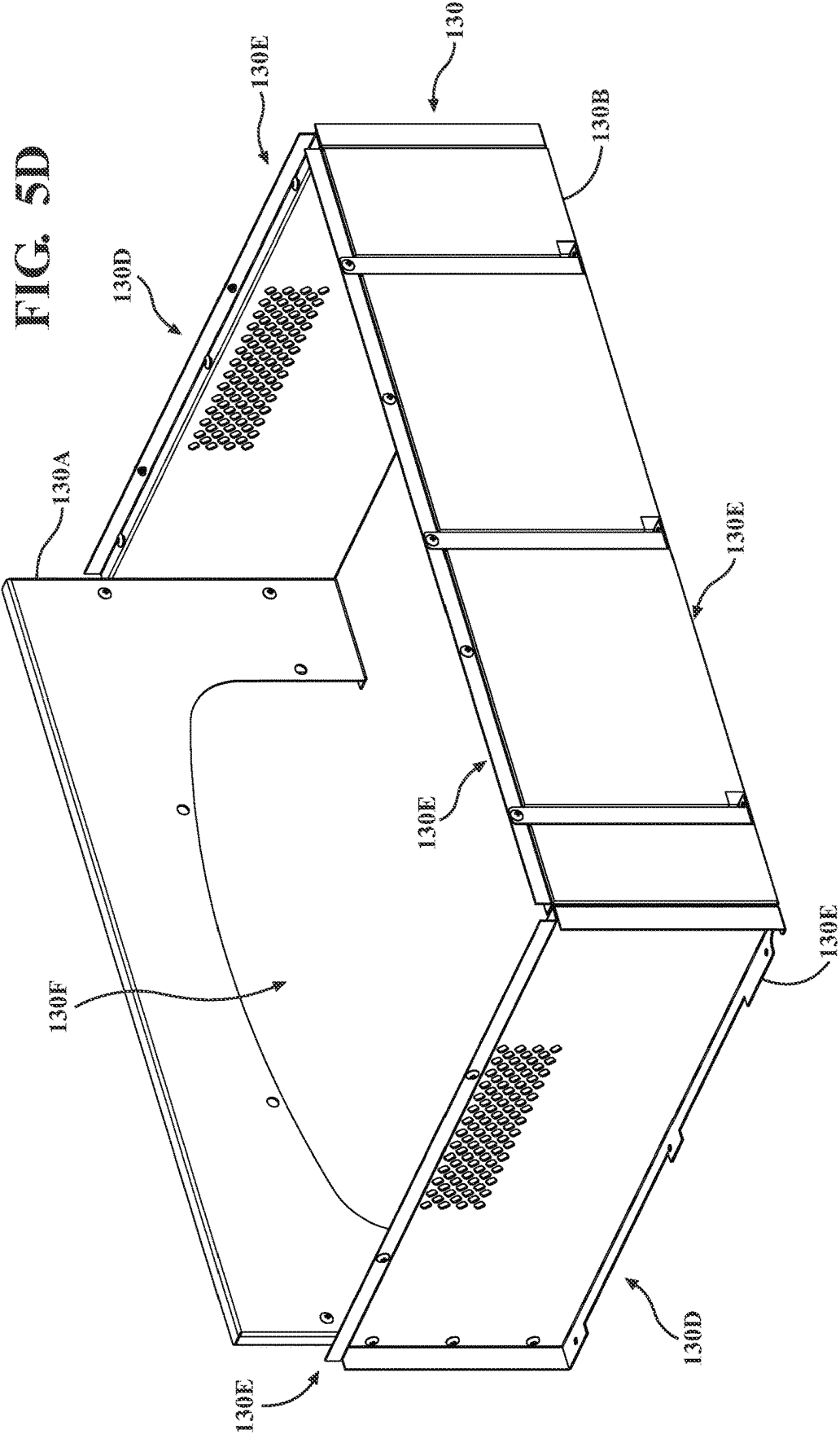


FIG. 5C





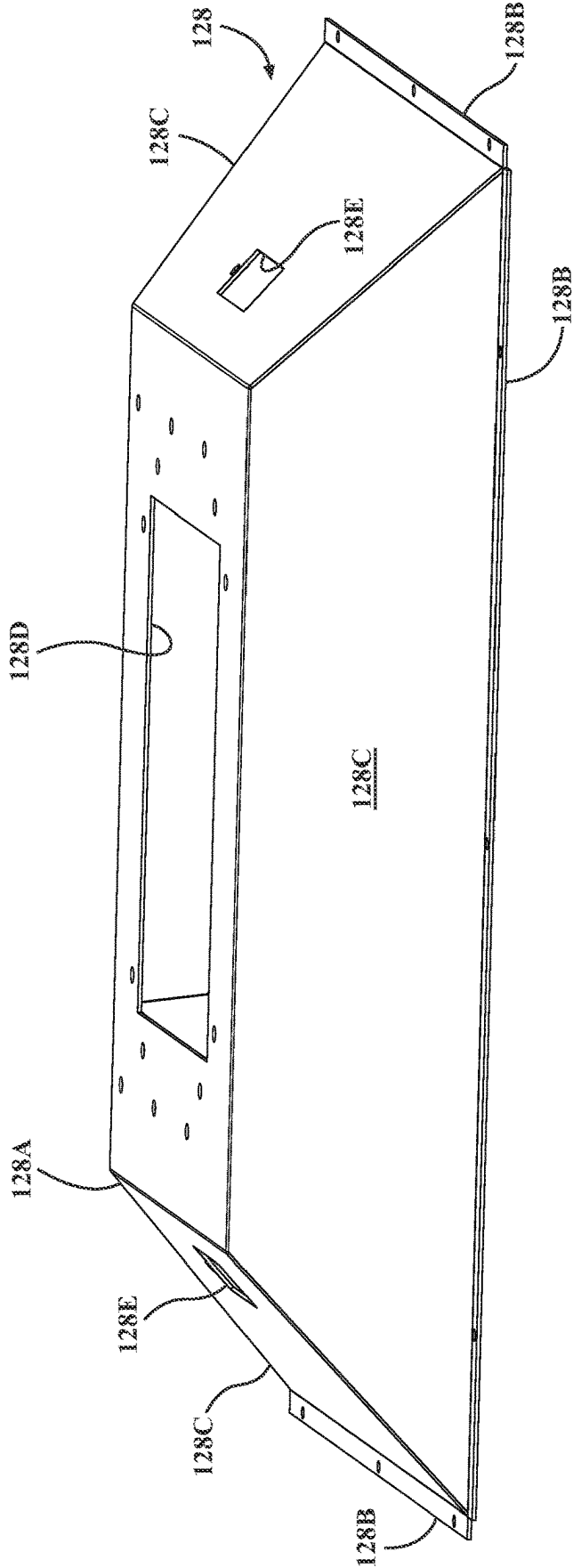
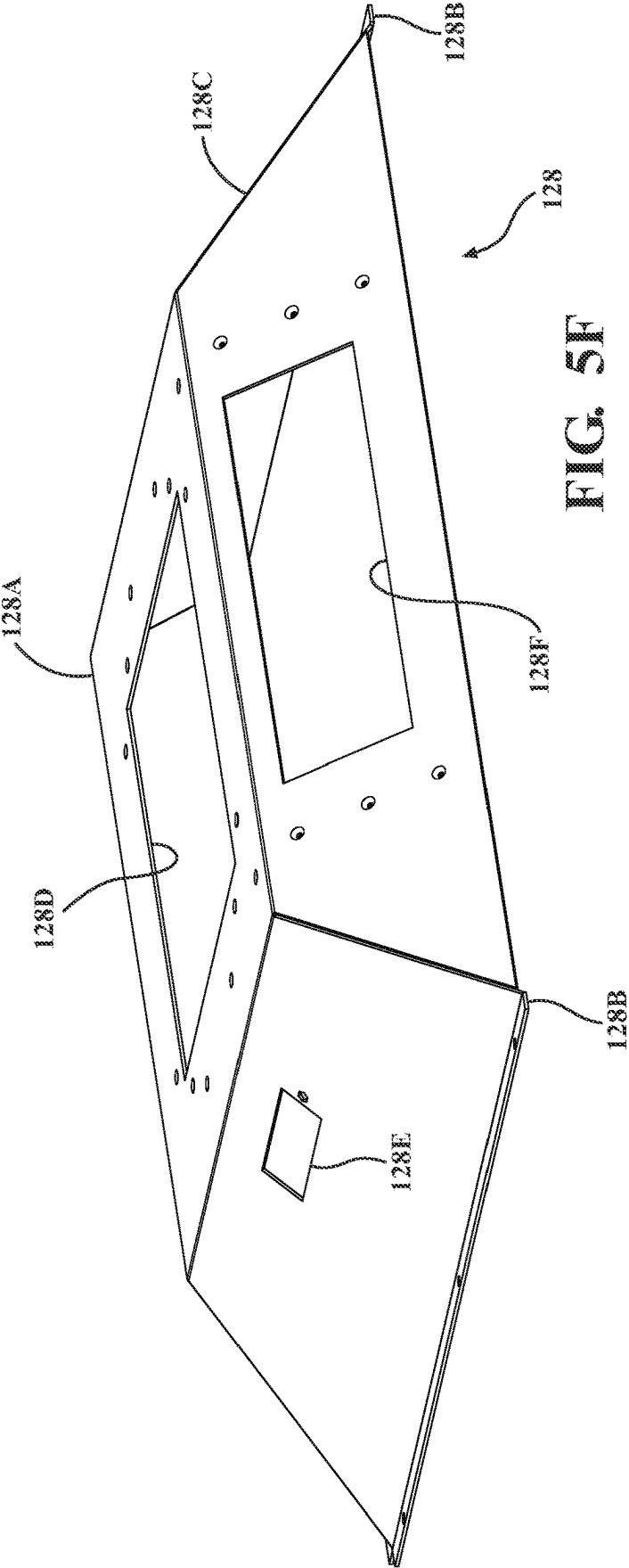
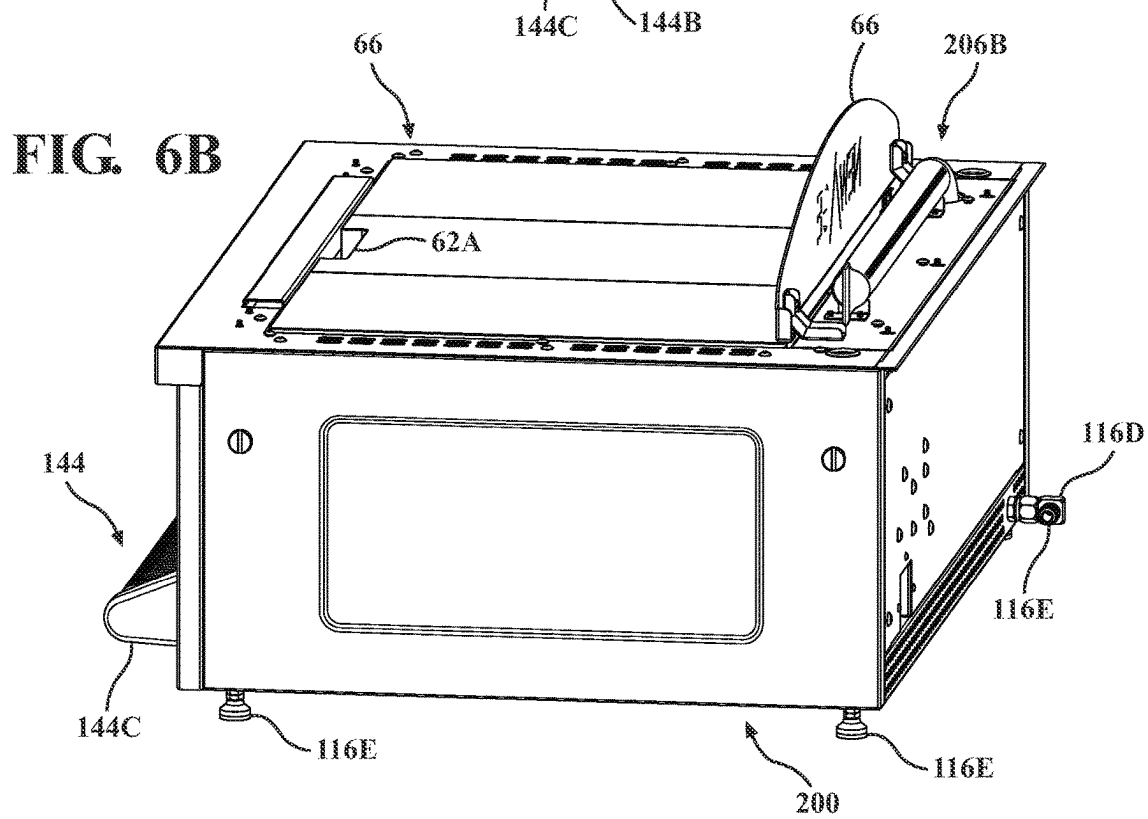
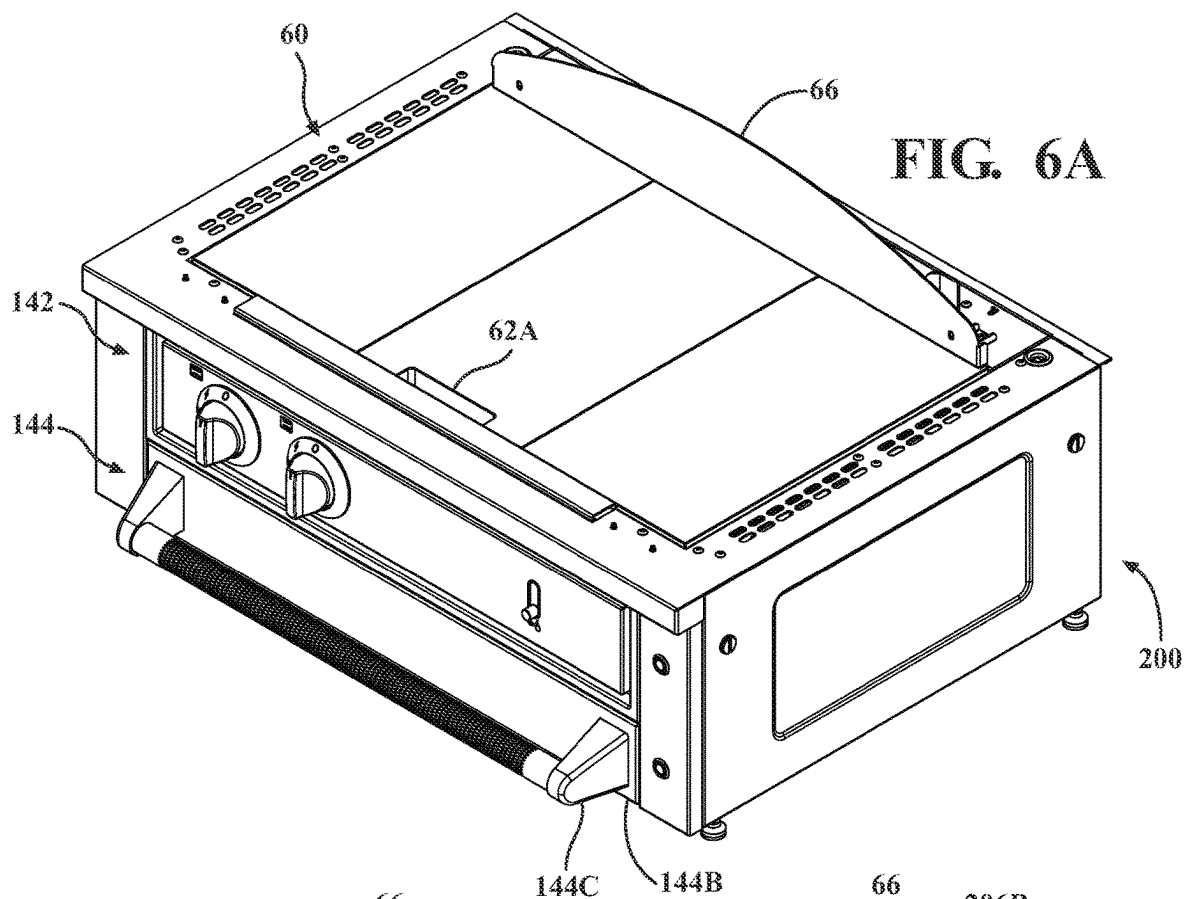


FIG. 5E





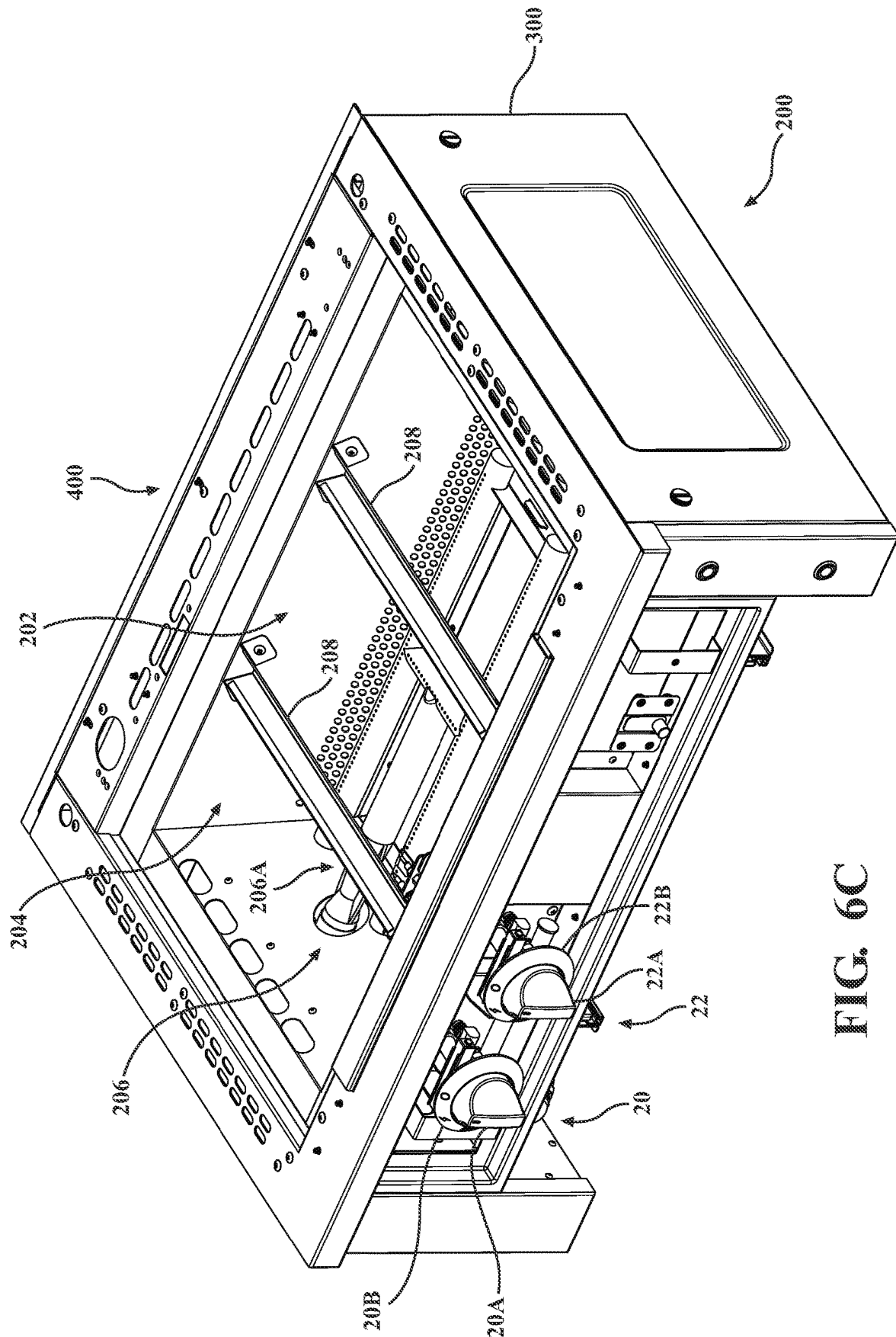


FIG. 6

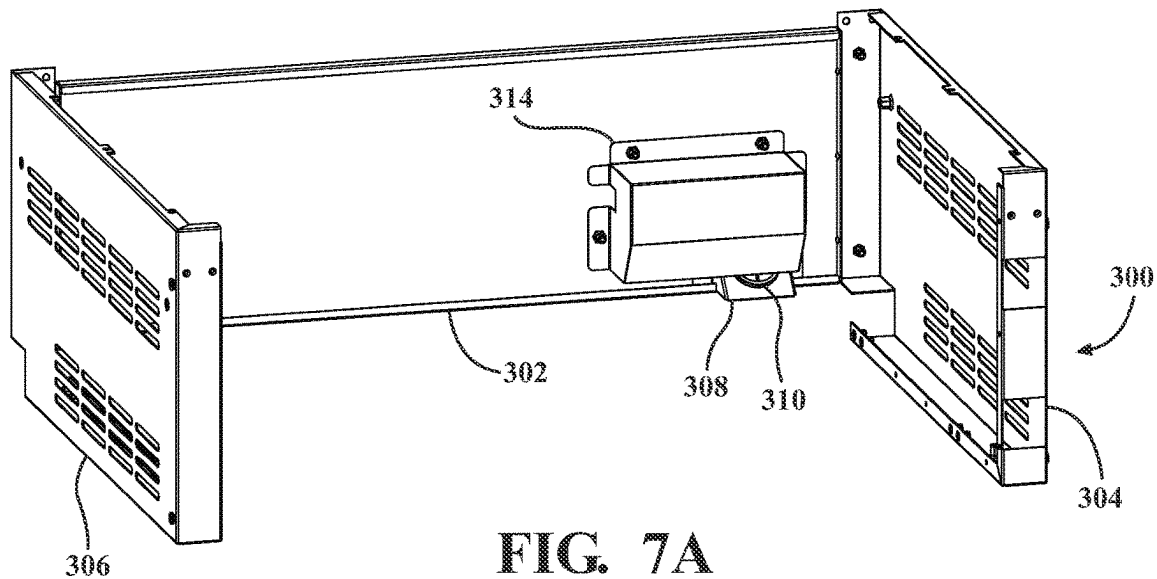


FIG. 7A

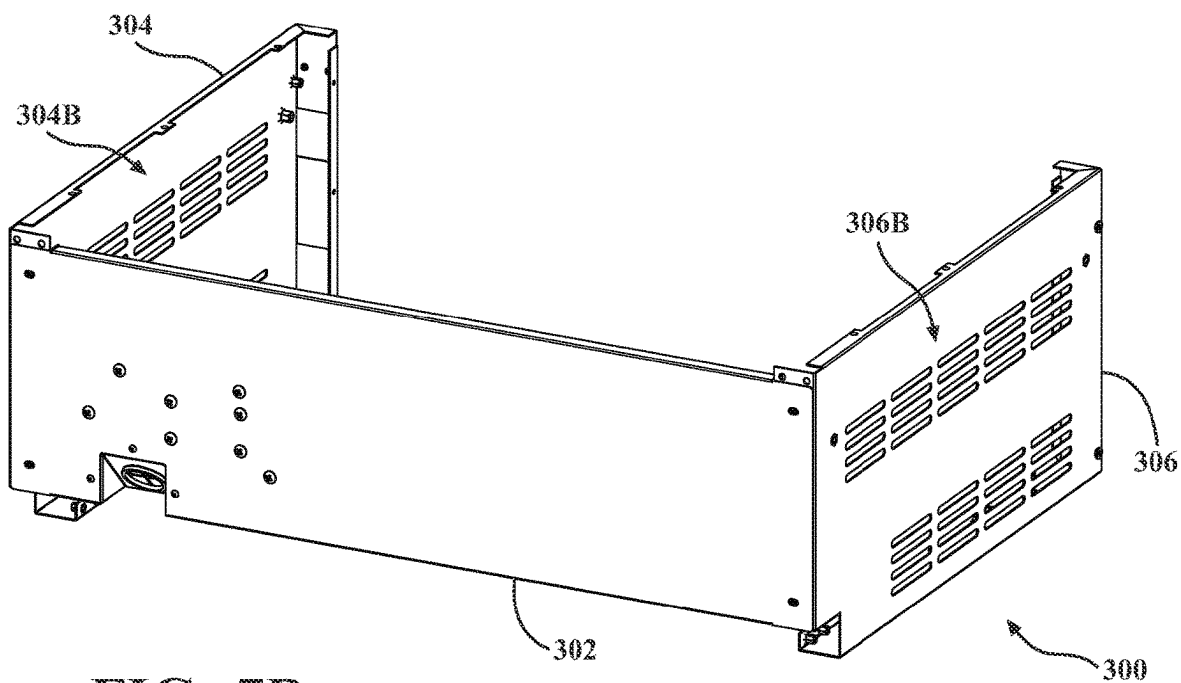
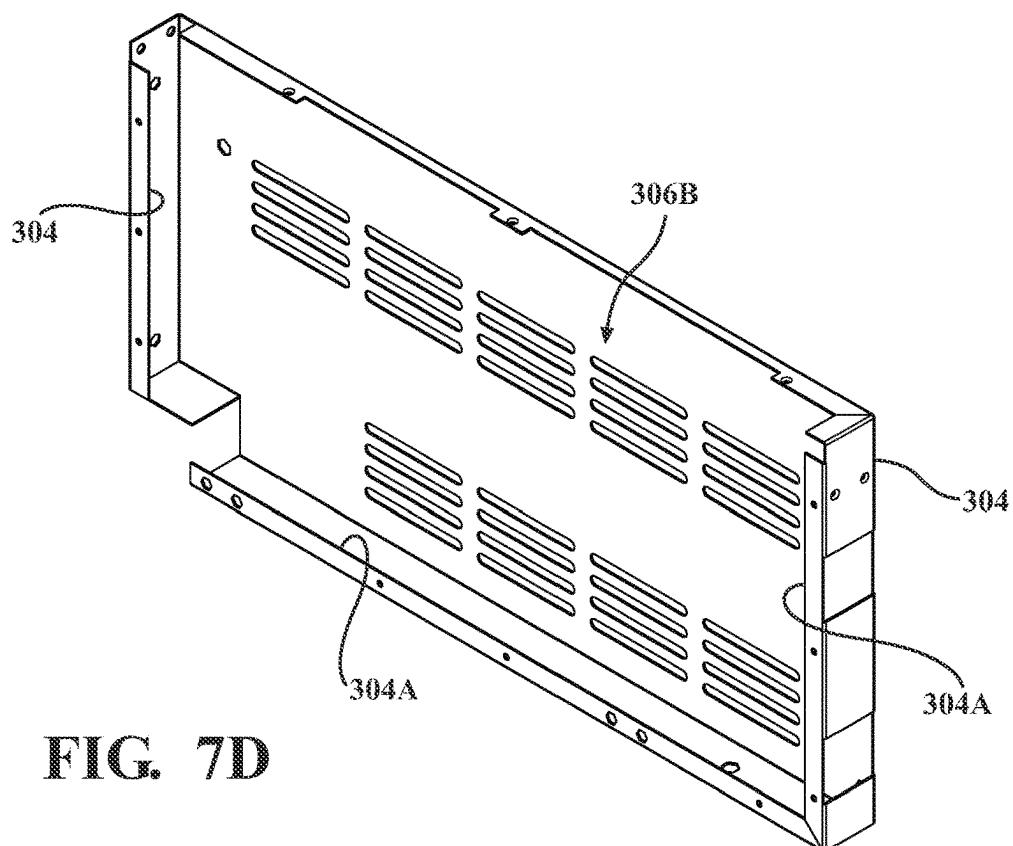
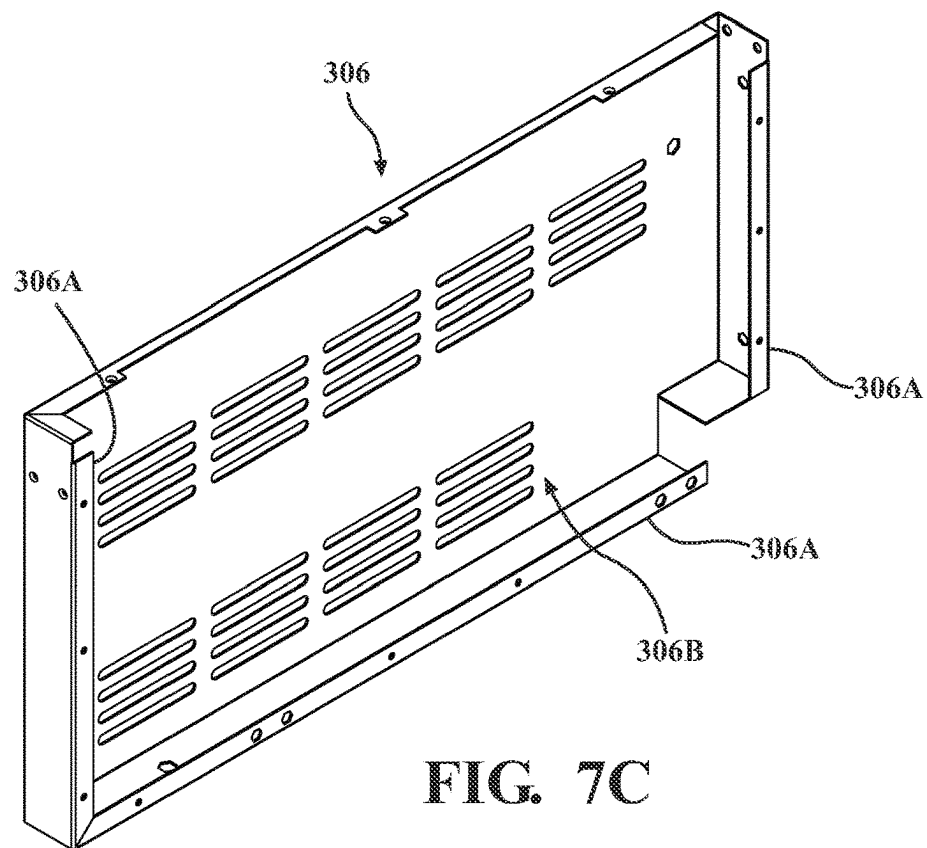
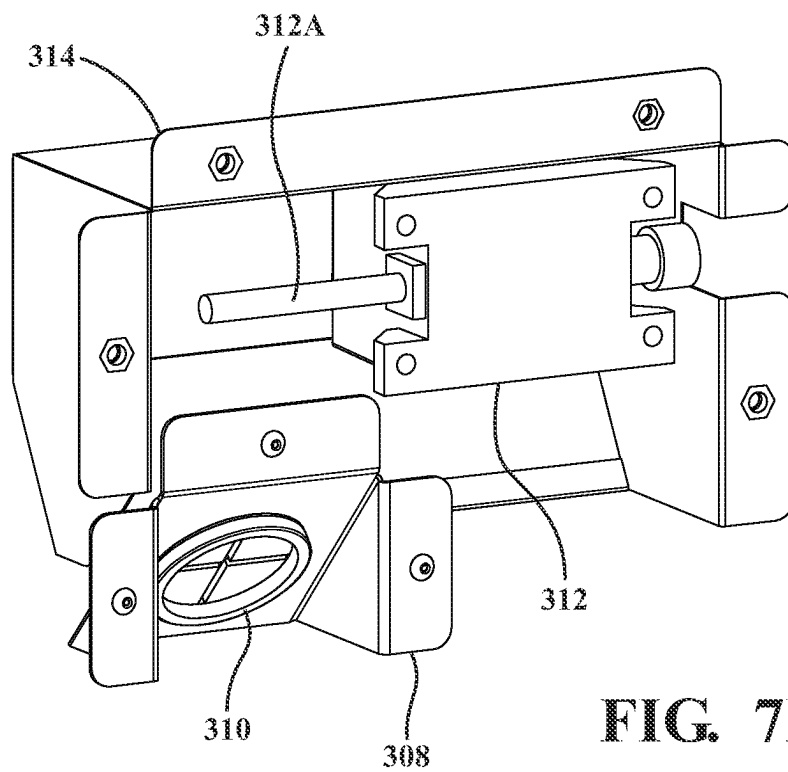
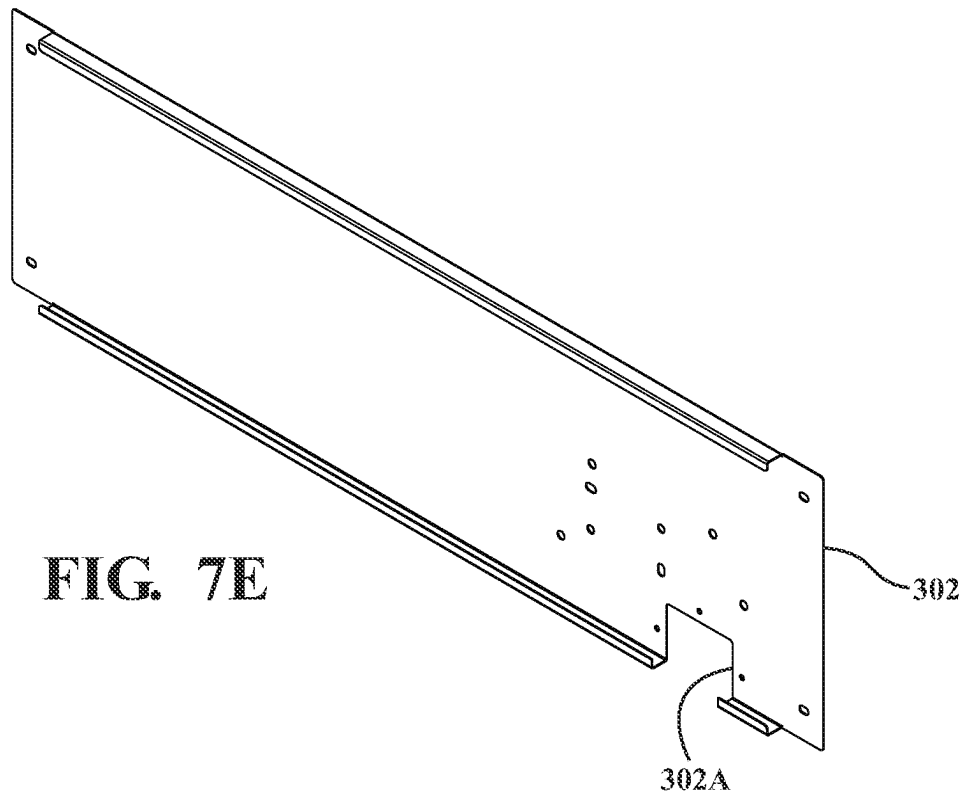
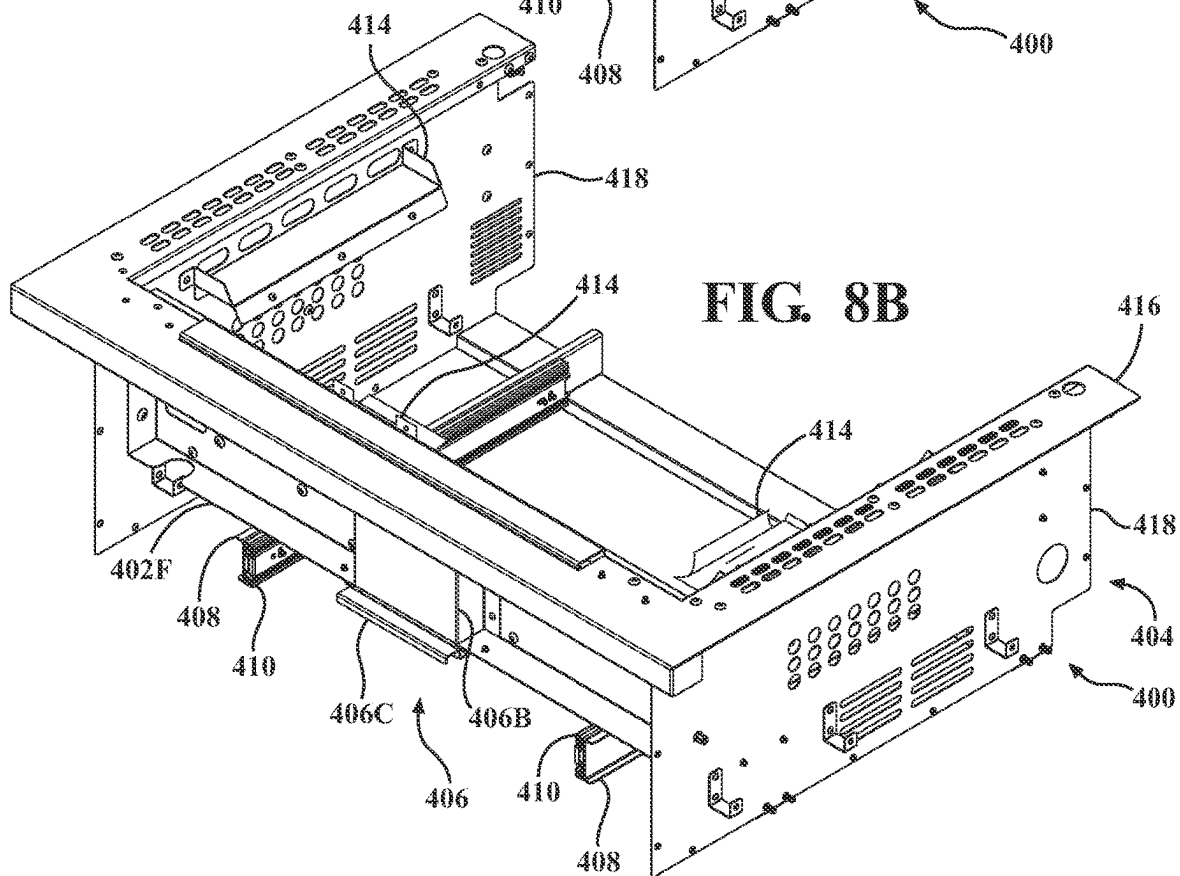
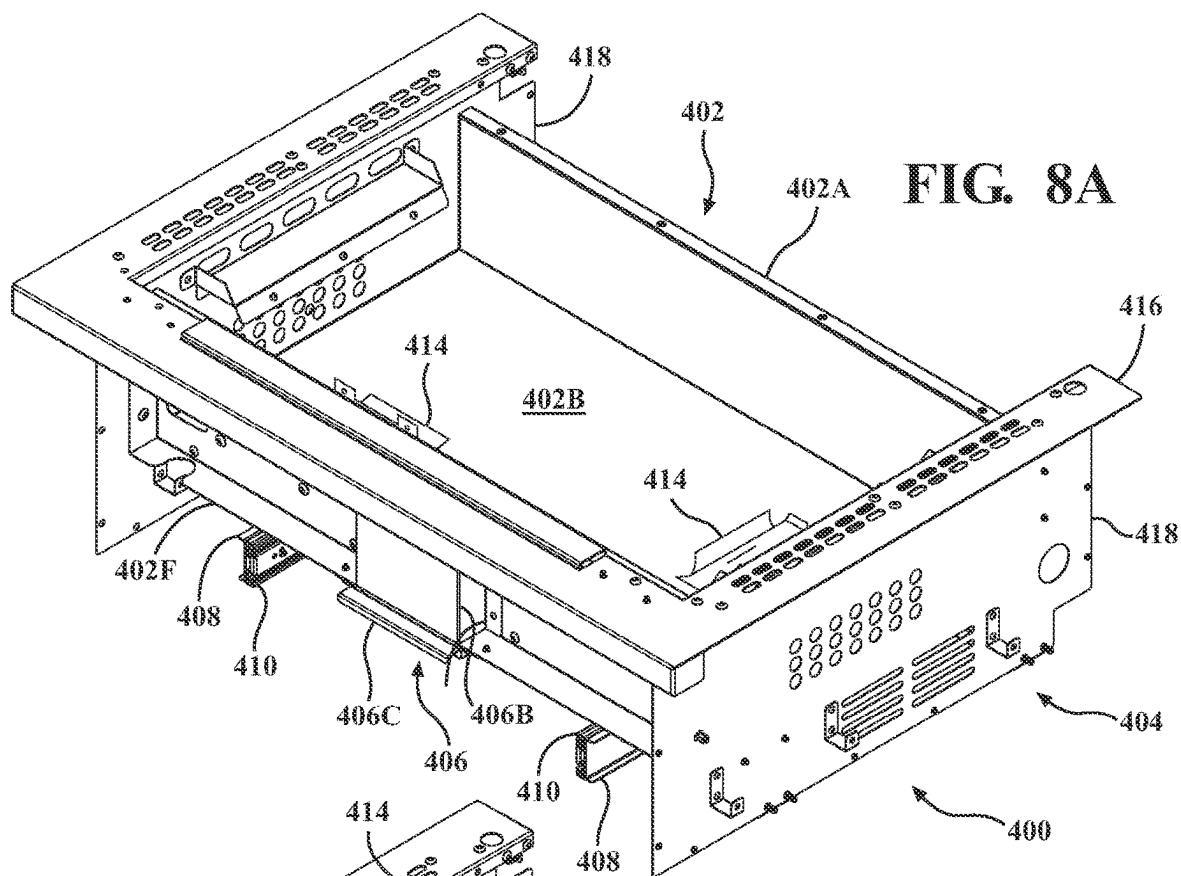


FIG. 7B







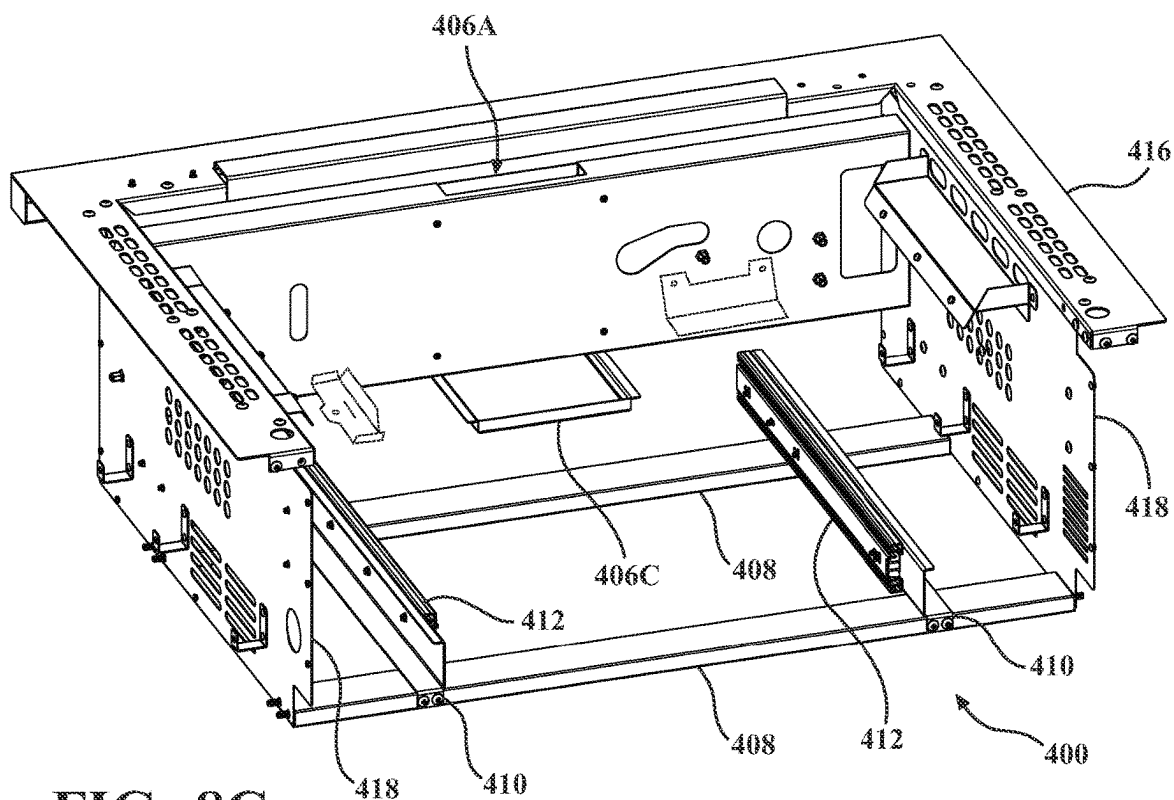


FIG. 8C

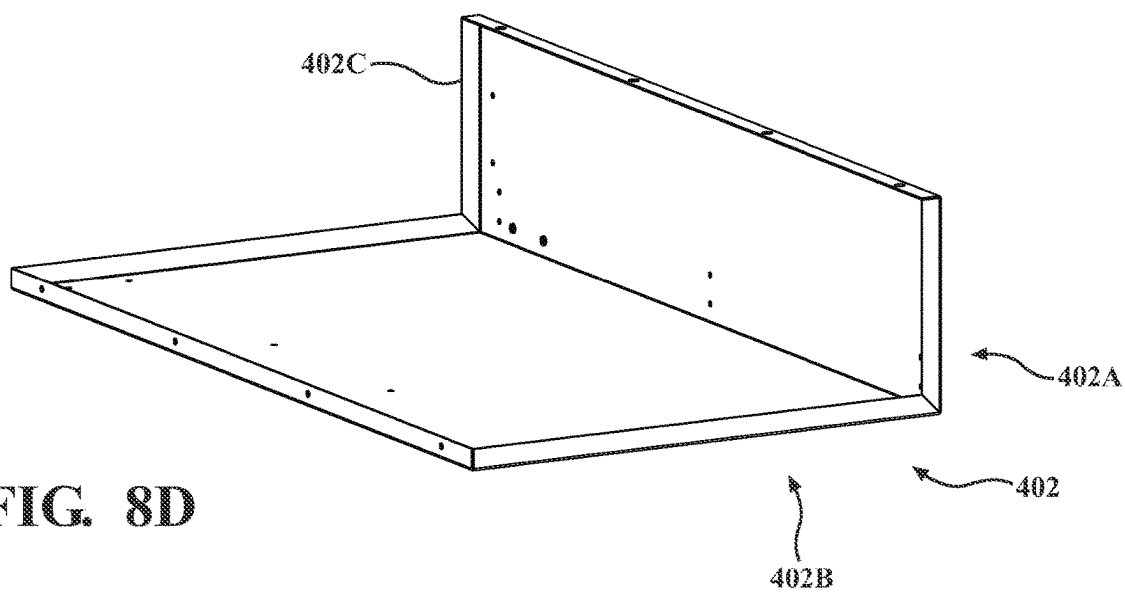
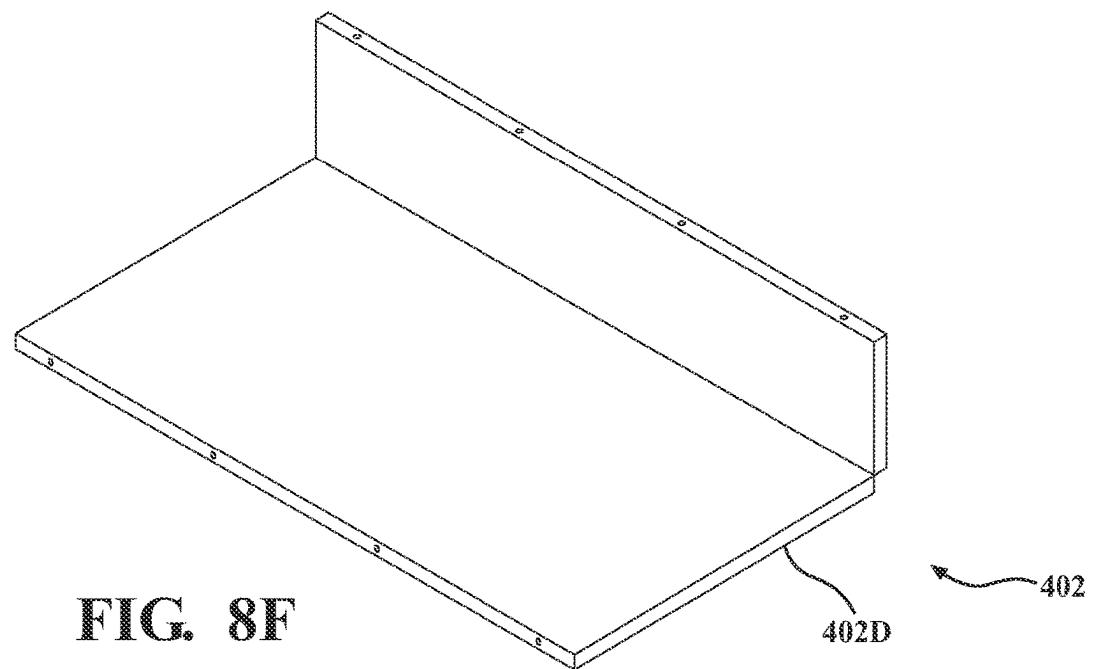
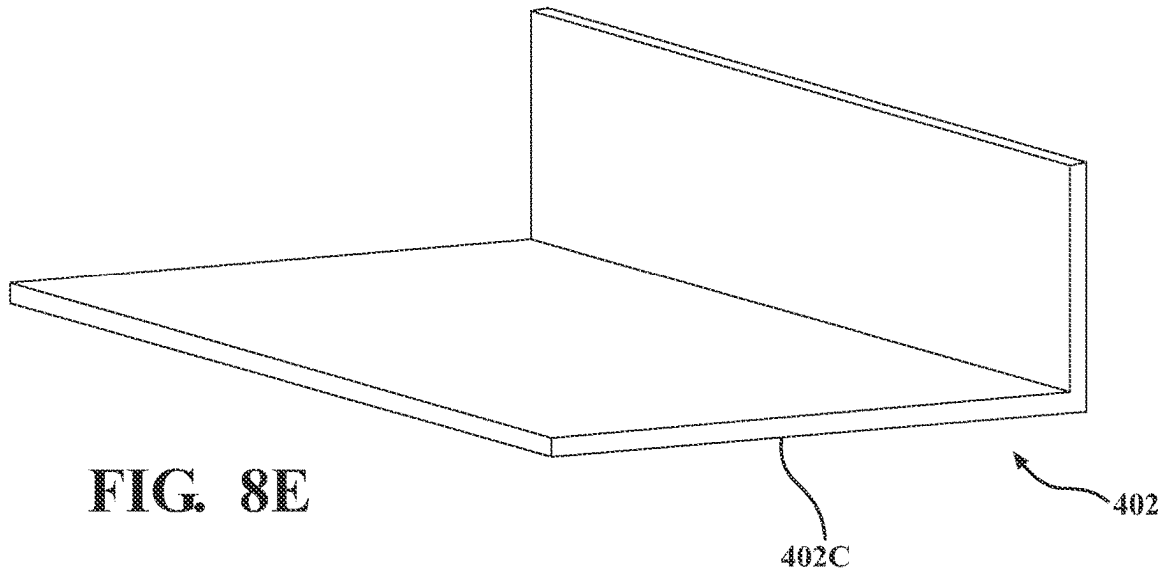


FIG. 8D



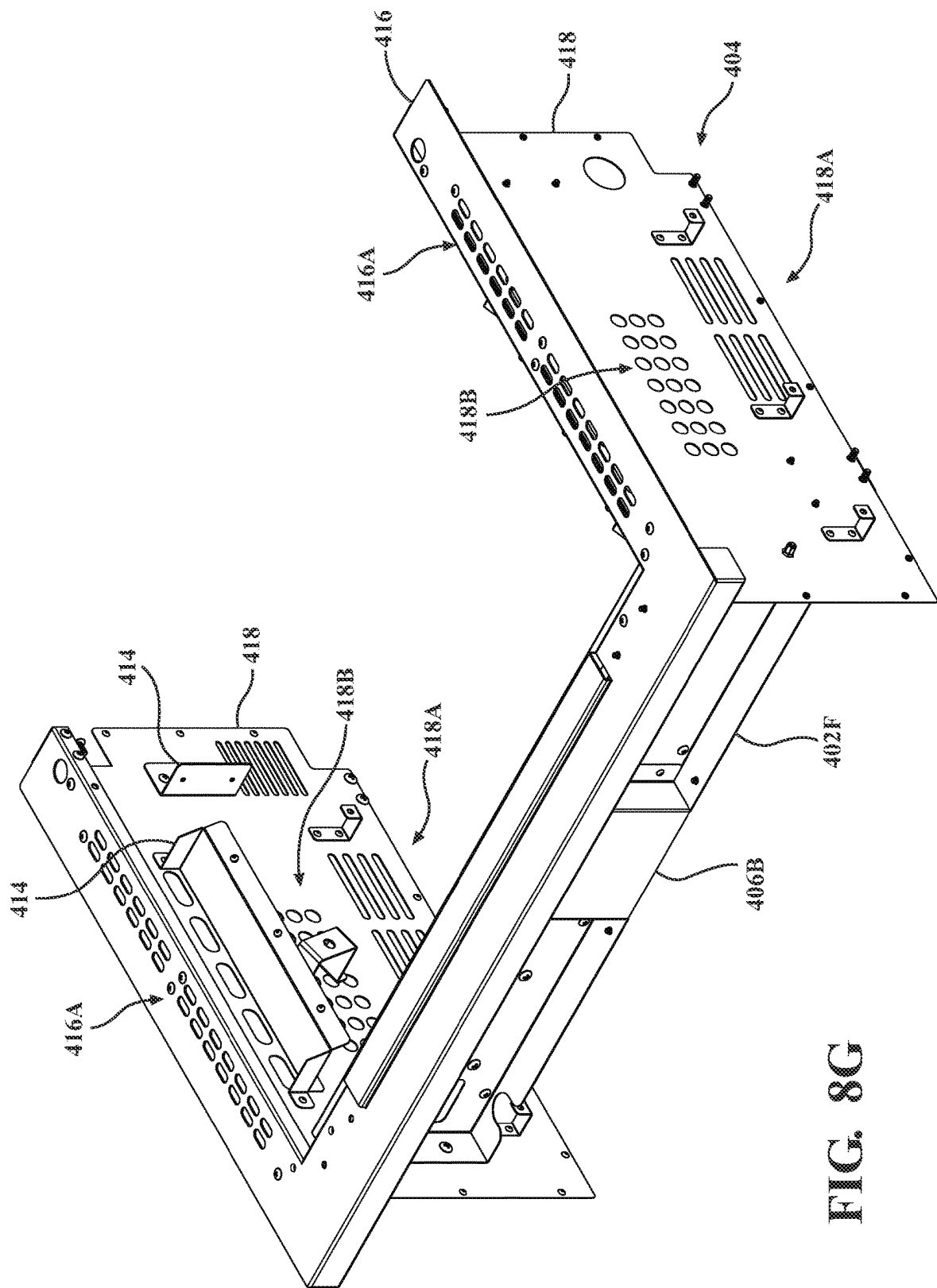


FIG. 8G

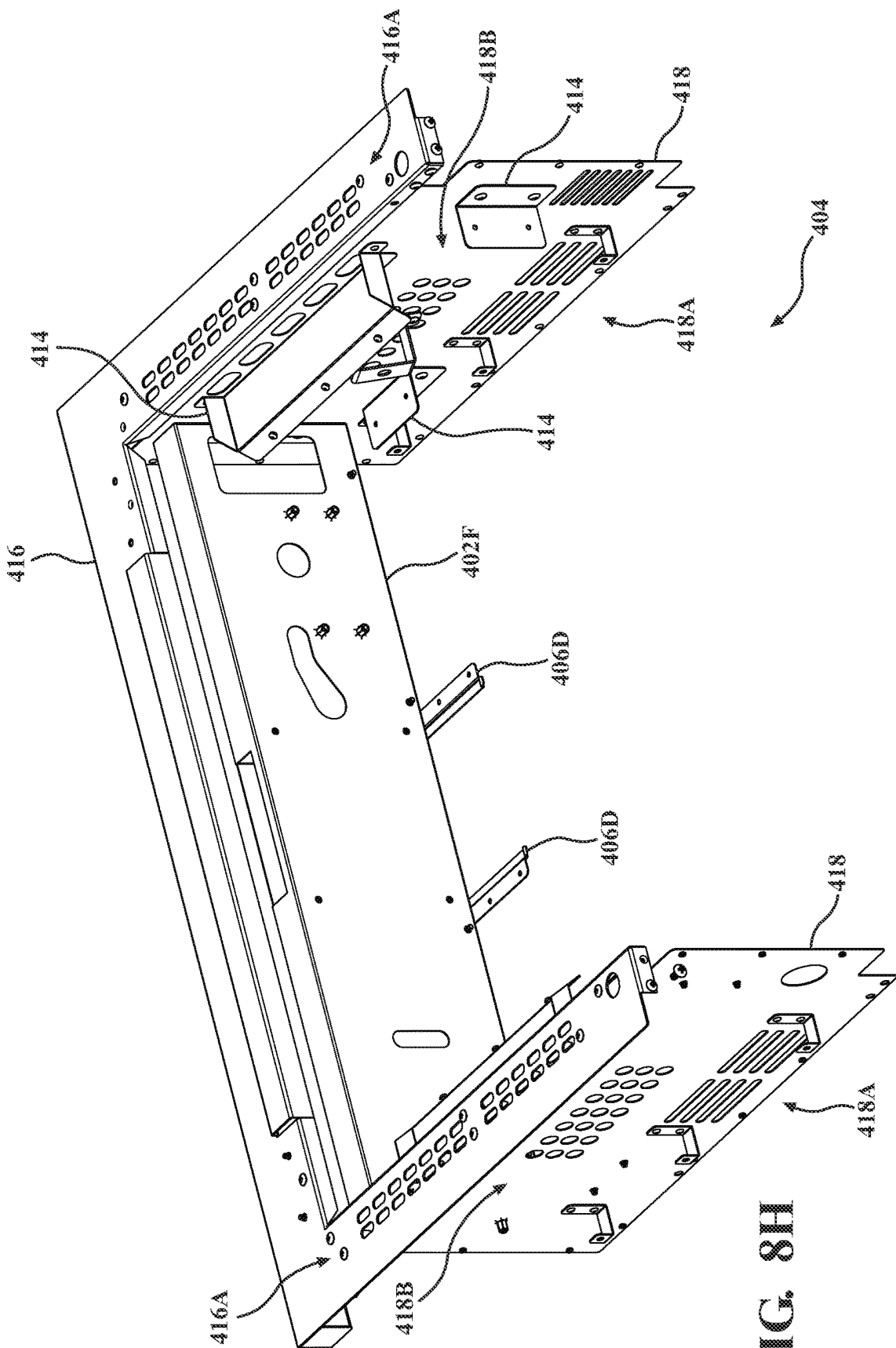


FIG. 8H

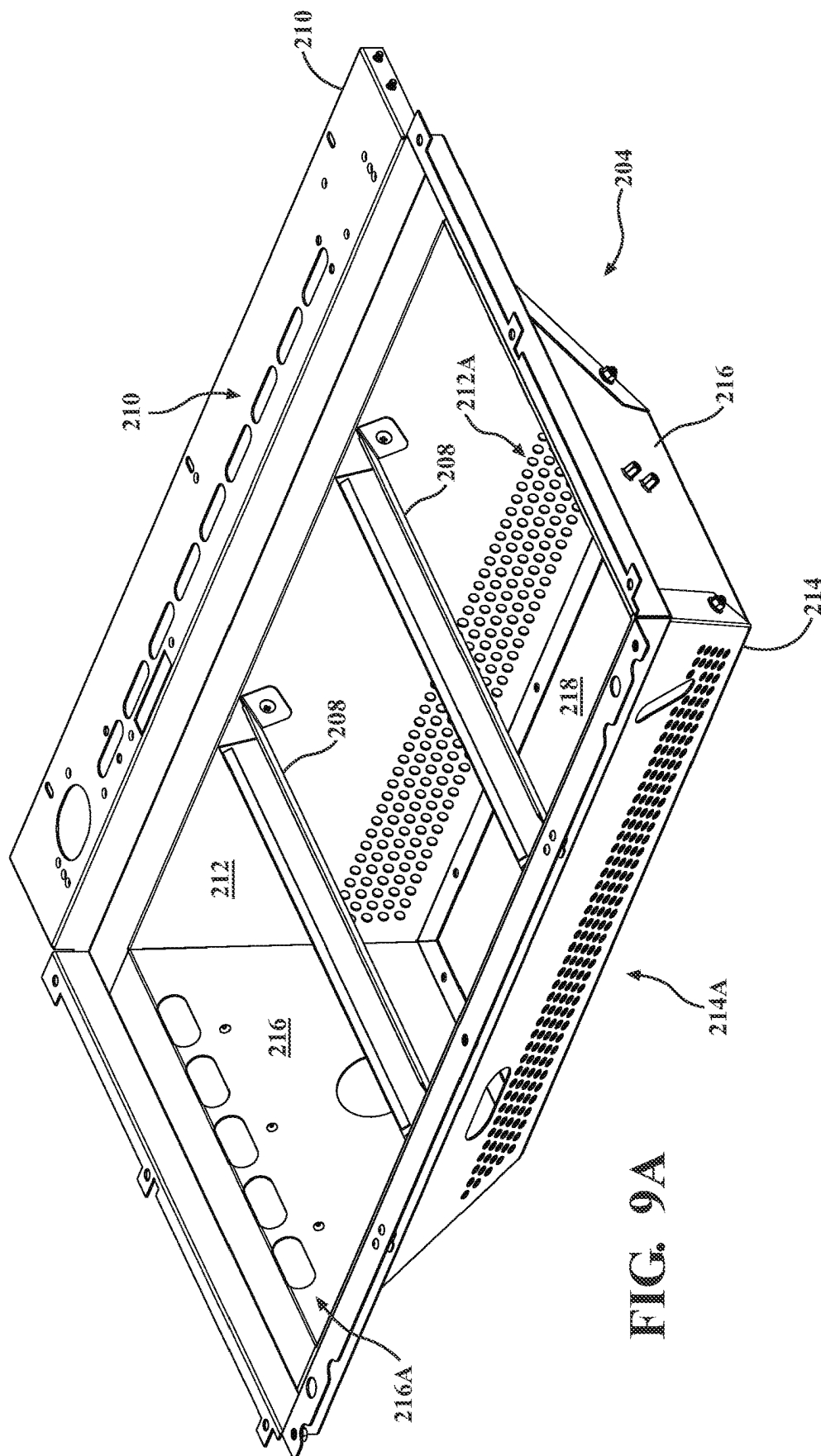
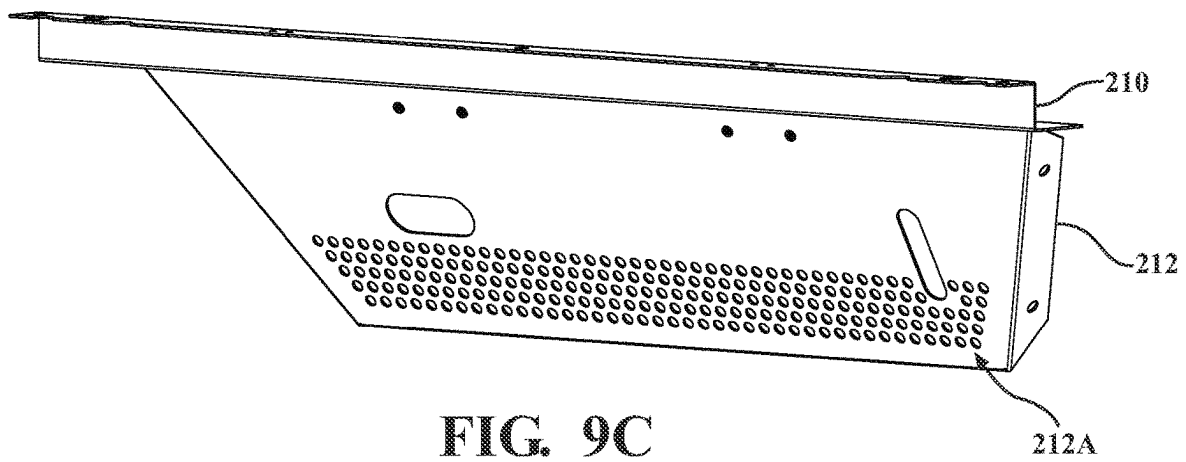
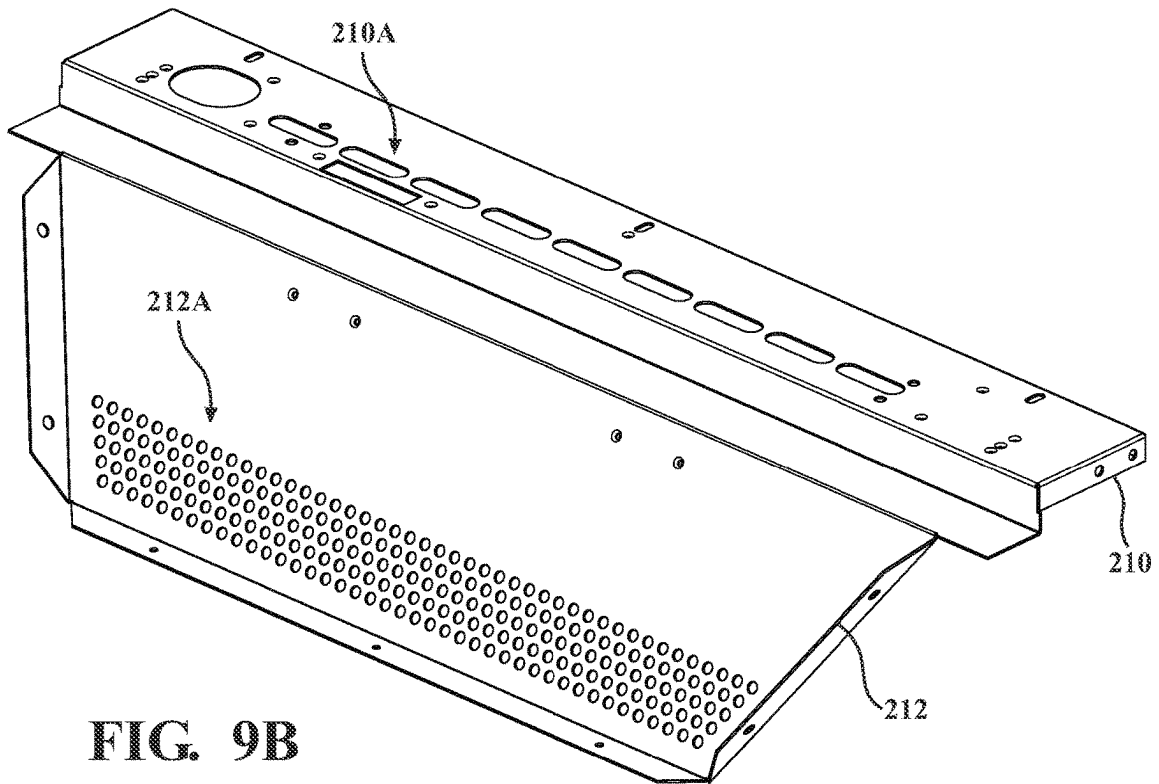
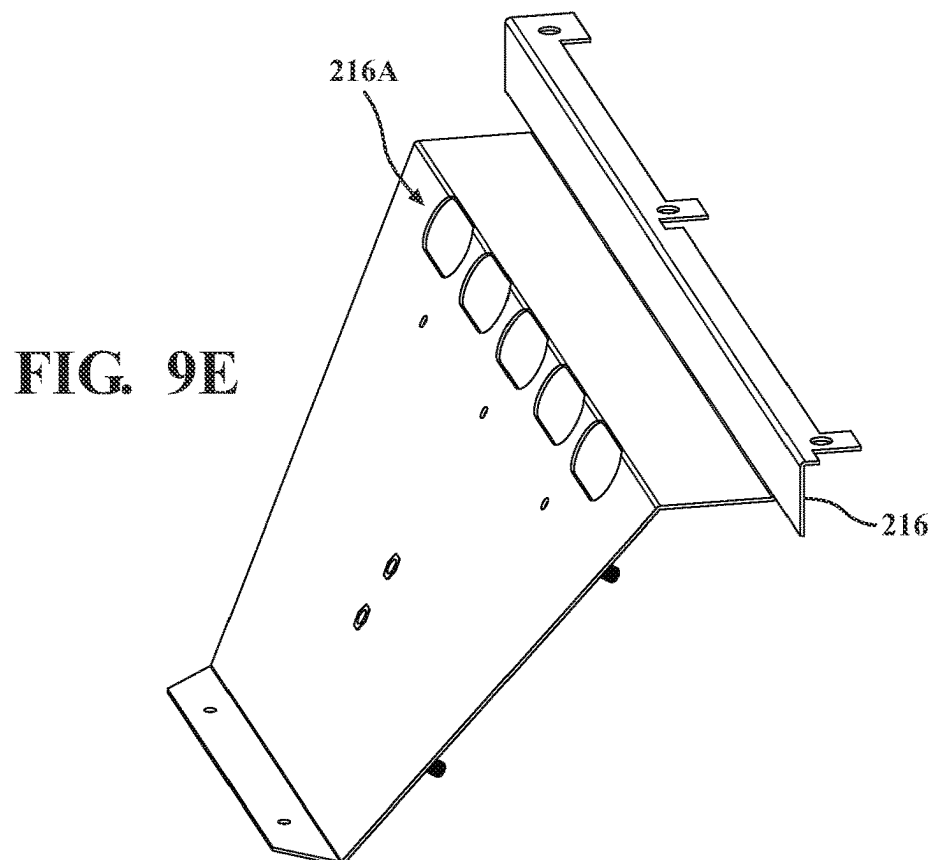
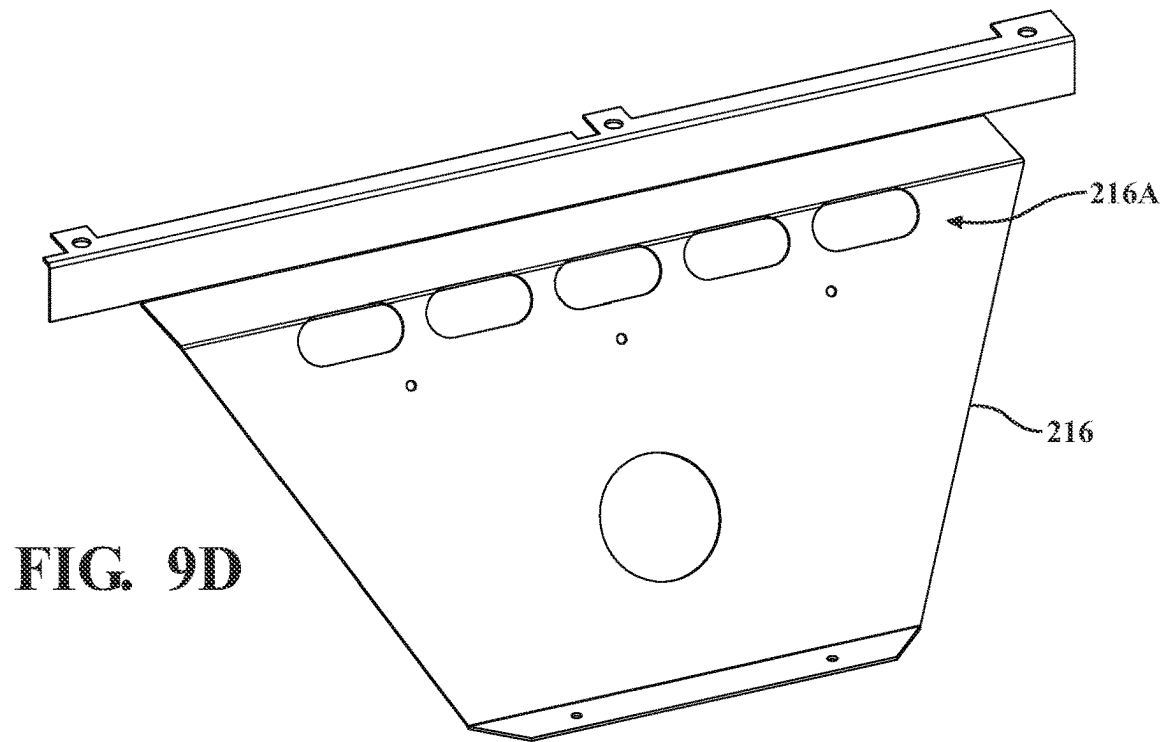
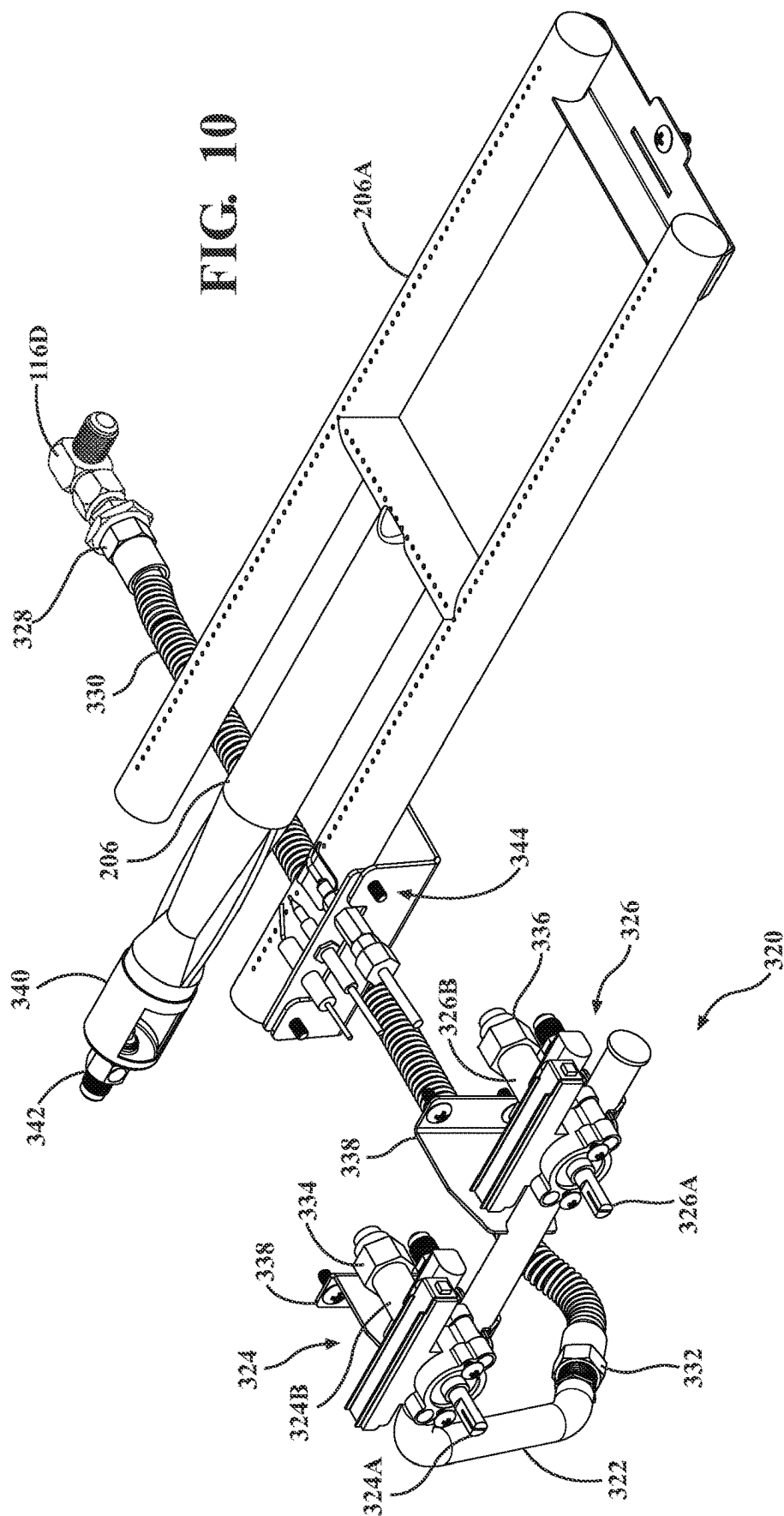
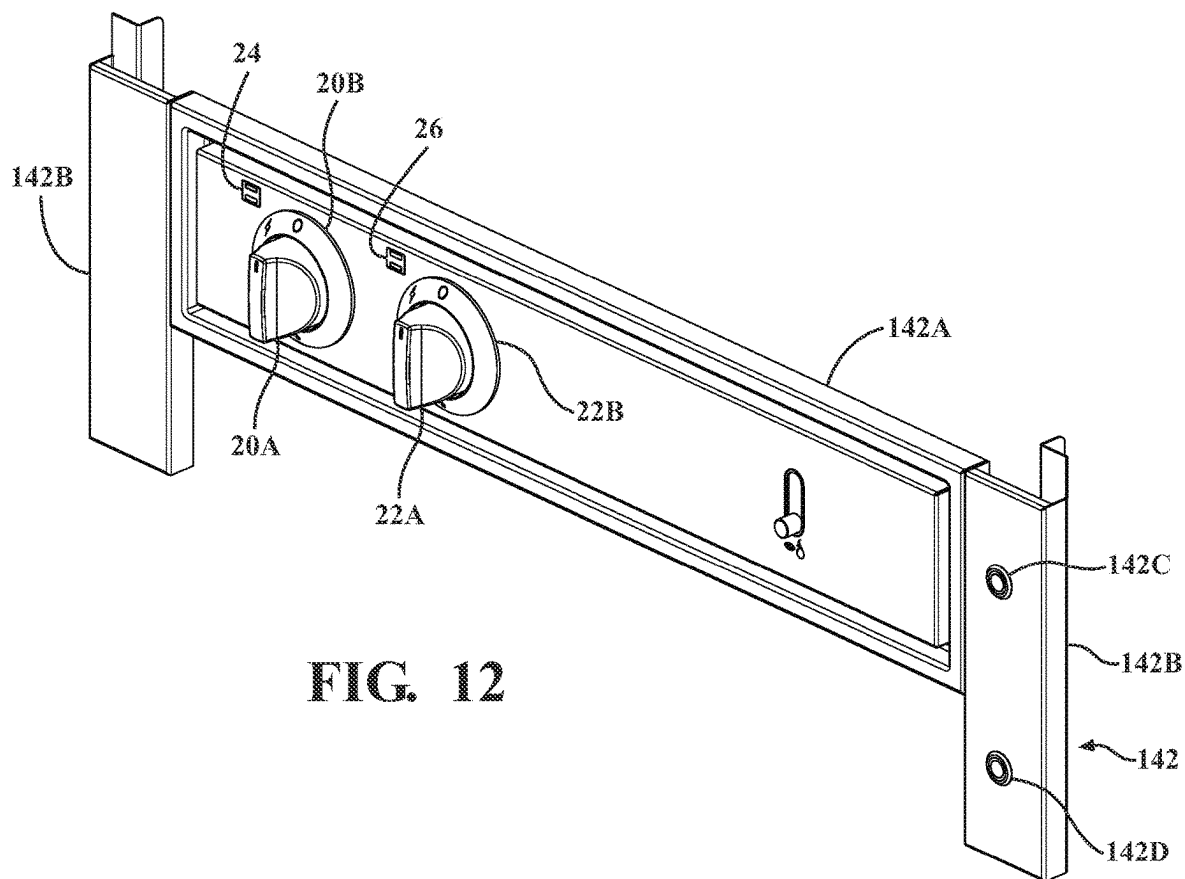
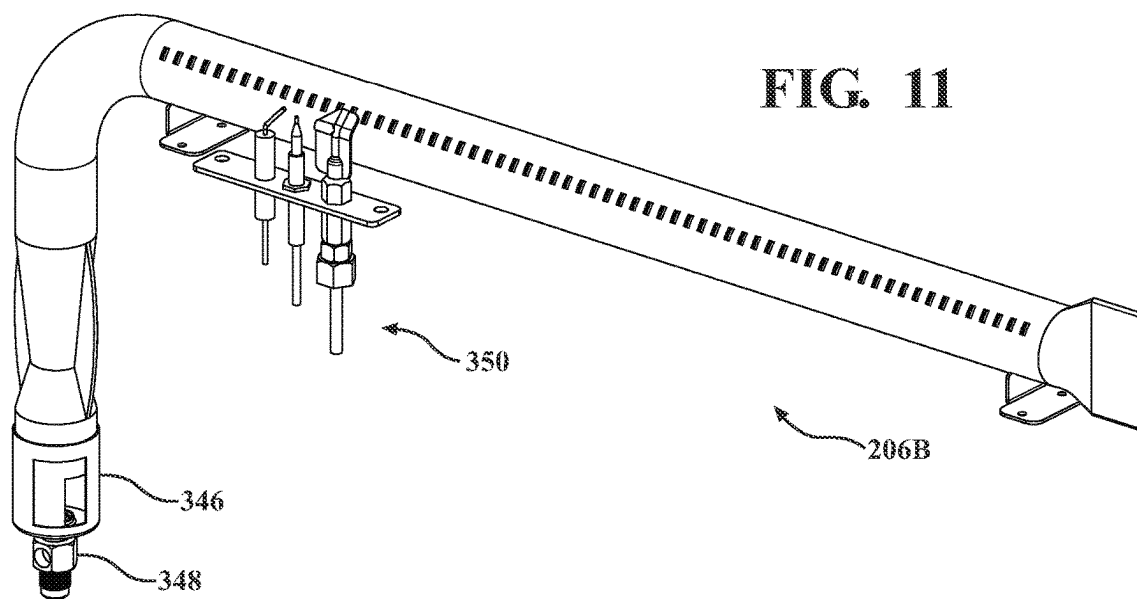


FIG. 9A









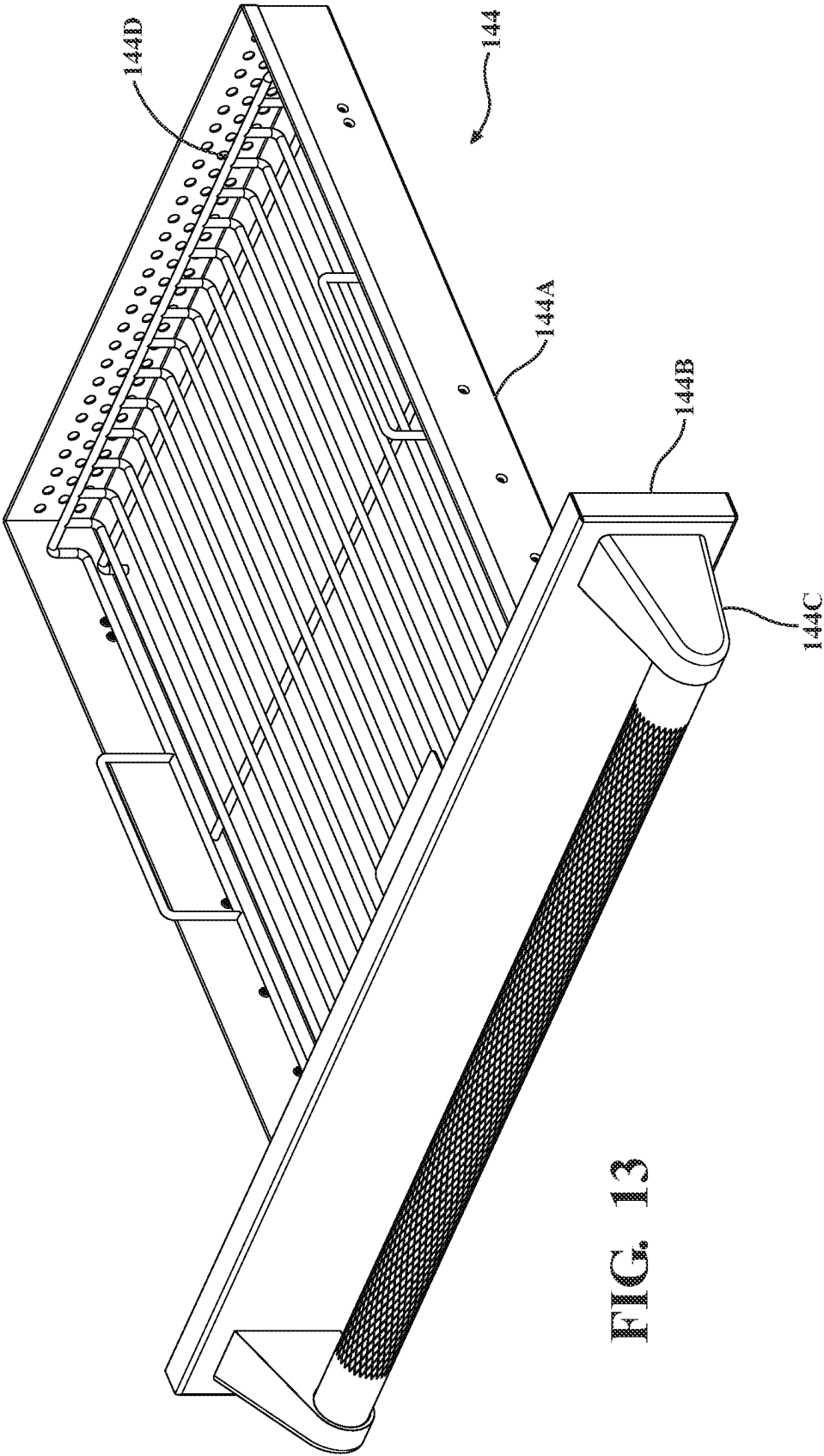


FIG. 13

1 OVEN

FIELD OF THE INVENTION

The present invention relates to ovens, and more specifically, to an oven having a lower assembly with a fire box, a cooking chamber and an upper assembly with two airflow pathways connected to the fire box.

BACKGROUND OF THE INVENTION

Many people enjoy cooking, indoor or outdoor, to prepare or cook a variety of foods, for example, pizza. Ovens that are designed or adapted specifically for cooking pizzas are widely popular. Generally, such ovens are capable of reaching very high temperatures and cooking a pizza within minutes. However, traditional pizza ovens tend to be large, cumbersome and inefficient.

The present invention is aimed at one or more of the problems identified above.

SUMMARY OF THE INVENTION

In one aspect of the present invention, an oven having upper and lower assemblies, a firebox and a heat absorbing and emanating device is provided. The upper assembly forms a cooking chamber and has a front side with an open entry to the cooking chamber. The lower assembly is located below the upper assembly and has an internal cavity. The firebox is located within the internal cavity of the lower assembly and includes a heat source. The heat absorbing and emanating device is located between the upper and lower assemblies and is configured to absorb heat from the heat source and to emanate heat into the cooking chamber of the upper assembly. The upper assembly includes outer and inner dome assemblies. The inner dome is located within the outer dome and forms first and second airflow pathways, a second airflow pathway, each of the first and second airflow pathways having an open end located at a bottom of the upper assembly and being open to the internal cavity of the lower assembly, the first airflow pathway having a second end terminating at a heating vent located on an inner wall of the inner dome and configured to allow heated air from the heat source to enter the cooking chamber, the second airflow pathway having a second end located terminating in an upper portion of the upper assembly.

In a second aspect of the present invention, an oven having upper and lower assemblies, a firebox, first and second heat sources, a heat guard, a thermometer, at least one removable door and heat absorbing and emanating device. The upper assembly has an upper portion and a bottom. The upper assembly forms a cooking chamber and has a front side with an open entry to the cooking chamber. The lower assembly is located below the upper assembly with the lower assembly having an internal cavity. The firebox is located within the internal cavity of the lower assembly. The first heat source is located within the firebox. The second heat source is located within the upper assembly for heating the cooking chamber. The heat guard is located between the second heat source and the cooking chamber. The thermometer is coupled to the lower assembly for sensing a temperature associated with the cooking chamber. The at least one removable door is rotatably coupled to the upper assembly within the open entry. The heat absorbing and emanating device is located between the upper and lower assemblies and is configured to absorb heat from the heat source and emanate heat into the cooking chamber of

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the upper assembly. The upper assembly includes an outer dome and an inner dome with the inner dome being located within the outer dome. First and second airflow pathways are disposed between the inner and outer domes. Each of the first and second airflow pathways has an open end located at the bottom of the upper assembly and are open to the internal cavity of the lower assembly. The first airflow pathway has a second end terminating at a heating vent located on an inner wall of the inner dome and is configured to allow heated air from the heat source to enter the cooking chamber. The second airflow pathway has a second end terminating in the upper portion of the upper assembly. The upper assembly further includes at least one exhaust vent located within the upper portion of the upper assembly near the second end of the second airflow path for allowing heated gas to exit the oven. The outer dome forms an exterior of the upper assembly. The inner dome forms at least a portion of the cooking chamber. The first and second airflow pathways being located between the inner and outer domes.

BRIEF DESCRIPTION OF THE DRAWINGS

Other advantages of the present invention will be readily appreciated as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings wherein:

FIG. 1A is a perspective view of an oven in a tabletop configuration, having an upper assembly and a lower assembly, according to a first embodiment of the present invention;

FIG. 1B is a front elevational view of the oven shown in FIG. 1A;

FIG. 1C is a perspective view of the oven shown in FIG. 1A in an elevated configuration;

FIG. 1D is a perspective view of the oven shown in FIG. 1A in a built-in configuration;

FIG. 1E is a perspective view of an enclosure that may be used with the oven shown in FIG. 1A;

FIG. 1F is a perspective view of a thermometer of the oven of FIG. 1A;

FIG. 2A is a cross-sectional view of the oven shown in FIG. 1A;

FIG. 2B is a cross-sectional view of the upper assembly of the oven shown in FIG. 1A;

FIG. 2C is a cross-sectional view of the lower assembly of the oven shown in FIG. 1A;

FIG. 3A is a first perspective view of an outer dome of the upper assembly of the oven of FIG. 1A;

FIG. 3B is a second perspective view of the outer dome of FIG. 3A;

FIG. 3C is a first perspective view of an outer dome top panel of the outer dome of FIG. 3A;

FIG. 3D is a second perspective view of the outer dome top panel of FIG. 3C;

FIG. 3E is a third perspective view of the outer dome top panel of FIG. 3C;

FIG. 3F is a first perspective view of an outer dome front panel of the outer dome of FIG. 3A;

FIG. 3G is a second perspective view of the outer dome front panel of FIG. 3F;

FIG. 3H is a perspective view of an outer dome back panel of the outer dome of FIG. 3A;

FIG. 3I is a perspective view of an outer dome side panel of the outer dome of FIG. 3A;

FIG. 4A is a first perspective view of a first door of the oven of FIG. 1A;

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FIG. 4B is a second perspective view of the first door of FIG. 4A;

FIG. 4C is a first perspective view of an outer door panel of the first door of FIG. 4A;

FIG. 4D is a second perspective view of the outer door panel of FIG. 4C;

FIG. 4E is a perspective view of a plurality of door brackets of the first door of FIG. 4A;

FIG. 4F is a perspective view of a plurality of insulation panels of the first door of FIG. 4A;

FIG. 4G is a perspective view of a hinge for the first door;

FIG. 5A is a first perspective view of an inner dome, having an inner dome upper portion and an inner dome lower portion, according to an embodiment of the present invention;

FIG. 5B is a second perspective view of the inner dome of FIG. 5A;

FIG. 5C is a first perspective view of the inner dome lower portion of FIG. 5A;

FIG. 5D is a second perspective view of the inner dome lower portion of FIG. 5A;

FIG. 5E is a first perspective view of the inner dome upper portion of FIG. 5A;

FIG. 5F is a second perspective view of the inner dome upper portion of FIG. 5A;

FIG. 6A is a first perspective view of the lower assembly of the oven of FIG. 1A;

FIG. 6B is a second perspective view of the lower assembly of the oven of FIG. 1A;

FIG. 6C is a third perspective view of the lower assembly of the oven of FIG. 1A;

FIG. 7A is a first perspective view of an outer enclosure of the lower assembly of the oven of FIG. 1A;

FIG. 7B is a second perspective view of the outer enclosure of FIG. 7A;

FIG. 7C is a perspective view of a first side panel of the outer enclosure of FIG. 7A;

FIG. 7D is a perspective view of a second side panel of the outer enclosure of FIG. 7A;

FIG. 7E is a perspective view of a back panel of the outer enclosure of FIG. 7A;

FIG. 7F is a perspective view of a transformer assembly of the outer enclosure of FIG. 7A;

FIG. 8A is a first perspective view of an outer enclosure of the lower assembly of the oven of FIG. 1A;

FIG. 8B is a second perspective view of the outer enclosure of FIG. 8A;

FIG. 8C is a third perspective view of the outer enclosure of FIG. 8A;

FIG. 8D is a perspective view of a heat shield bottom panel of a heat shield of the outer enclosure of FIG. 8A;

FIG. 8E is a perspective view of a heat shield insulation panel of the heat shield;

FIG. 8F is a perspective view of a heat shield bottom panel of the heat shield;

FIG. 8G is a first perspective view of a fire box support structure of the outer assembly of FIG. 8A;

FIG. 8H is a second perspective of the fire box support structure of FIG. 8G;

FIG. 9A is a perspective view of a fire box of the oven of FIG. 1A;

FIG. 9B is a perspective view of a fire box back panel of the fire box of the FIG. 9A;

FIG. 9C is a perspective view of a fire box front panel of the fire box of FIG. 9A;

FIG. 9D is a perspective view of a first fire box side panel of the fire box of the FIG. 9A;

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FIG. 9E is a perspective view of a second fire box side panel of the fire box of FIG. 9A;

FIG. 10 is a perspective view of a main burner of the oven of FIG. 1A;

FIG. 11 is a perspective view of a rear burner of the oven of FIG. 1A;

FIG. 12 is a perspective view of a control panel assembly of the oven of FIG. 1A; and,

FIG. 13 is a perspective view of a drawer assembly of the oven of FIG. 1A.

DETAILED DESCRIPTION OF INVENTION

Referring to the FIGS., wherein like numerals indicate like or corresponding parts throughout the several views, an embodiment of an oven 10 having an upper assembly 100 and a lower assembly 200 for cooking items of food, such as pizza, as shown. With reference to FIGS. 2A-2C, the upper assembly 100 includes a cooking chamber 102 and an oven opening 104 through which the cooking chamber 102 may accessed.

Returning to FIGS. 1A-1D, in the illustrated embodiment, the oven 10 includes first and second doors 12A, 12B. As explained in further detail below, the first and second doors 12A, 12B may commercial, dual-layered doors and include a sealing gasket (see below) and are rotatably coupled to the upper assembly 100 via respectively hinges 14. The first and second doors 12A, 12B assist in quickly heating and maintaining the heat within the cooking chamber 102. In another aspect of the present invention, the first and second doors 12A, 12B may be removable. The oven 10 includes first and second handles 16 mounted to the respective first and second doors 12A, 12B. In one embodiment, the handles 16 may be rotated to lock/unlock the doors. In the illustrated embodiment, the first and second doors 12A, 12B are locked when the handles 16 are upright and parallel (as shown in FIG. 1B). The first and second doors 12A, 12B may be unlocked by rotating the handles 16 in opposite directions, as indicated by the arrows in FIG. 1B. A thermometer 18 measures and displays the temperature of the cooking chamber 102. The handles 16 may have a textured surface. For example, in the illustrated embodiment the surface of the handles 16 are knurled, i.e., have small ridges.

The oven 10 may be provided or arranged in difference configurations. For example as shown in FIGS. 1A and 1B, the oven 10 may be arranged in a tabletop configuration in which the lower assembly 200 of oven 10 is mounted inside an enclosure 116 and a lower edge or bottom of the lower assembly 200 is suspended a short distance over a resting surface, such as a non-combustible countertop (not shown). With particular reference to FIG. 1E, the enclosure 116 may include a first enclosure side panel 116A, a second enclosure side panel 116B and a back enclosure panel 116C which connects that first and second enclosure side panels 116A, 116B. Fuel or gas may be fed to the oven 10 via a swiveling elbow connector 116D. A plurality of swivel leveling mounts 116E are mounted to a bottom side of the first and second enclosure panels 116A, 116B. The swivel leveling mounts 116E are adjustable and allow the oven 10 to be leveled.

As shown, in FIG. 1C, the oven 10 may be arranged in an elevated configuration in which the lower assembly 200 is mounted within the enclosure 116 at an elevated or raised position. In the elevated configuration, the lower assembly 200 is mounted within the enclosure 116 such that the lower edge or bottom of the lower assembly 200 is elevated a

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greater distance above the resting surface (not shown) to provide additional airflow underneath the oven 10.

Additionally, the oven 10 may be provided or used without the enclosure 116 in a built-in configuration. The oven 10 in the built-in configuration may be mounted, for example, within other or existing indoor or outdoor cabinetry (not shown).

As shown in FIGS. 2A-2C, the lower assembly 200 includes an internal cavity 202 and a fire box 204 located within the internal cavity 202. The fire box 204 includes a first heat source 206. In one embodiment, the first heat source 206 includes a first (or main) gas burner 206A. In the illustrated embodiment, the first gas burner 206A is "H-shaped" and is rated at 18,000 BTU. A second (or back) heat source gas burner 206B (see below) extends above the lower assembly 200 into the cooking chamber 102 of the upper assembly 100. The second heat source 206B may be a rear gas tube burner rated at, e.g., 22,000 BTU.

The oven 10 further includes a first heat absorbing and emanating device 60 located between the upper and lower assemblies 100, 200. The first heat absorbing and emanating device 60 is configured to absorb heat from the heat source 206 and emanate heat into the cooking chamber 102 of the upper assembly 100. As shown in the illustrated embodiment, the first heat absorbing and emanating device 60 includes a plurality, e.g., three, of food grade ceramic cooking stones 62 that form a cooking surface 64. The ceramic cooking stones 62 may have a thickness of 30 mm. The plurality of ceramic cooking stones 62 are heated directed from below by the first gas burner 206A (as shown by arrows A) and emanate heat into the cooking chamber 102 to ensure that the bottom of the food product placed on the ceramic cooking stones 62 are heated thoroughly. The second gas burner 206B directly heats the cooking chamber 102 to cook the food product. A heat guard 66 separates the second gas burner 206B from the rest of the cooking chamber 102 of the upper assembly 100 to protect the food product from the flames from the second gas burner 206B.

In the illustrated embodiment, the oven 10 includes a second heat absorbing and emanating device 70 positioned at the top of the cooking chamber 102. As shown, includes a ceramic cooking stone 72.

As shown in FIGS. 2A-2C, the upper assembly 100 includes an outer dome 106 and an inner dome 108. The outer dome 106 forms the exterior of the upper assembly 100. The inner dome 108 is located within the outer dome 106 and forms a first airflow pathway 110 and a second airflow pathway 112. As shown, in the illustrated embodiment, each of the first and second airflow pathways 110, 112 having a first end 110A, 112A located at a bottom of the upper assembly 100 and being open to the internal cavity 202 of the lower assembly 200. The first airflow pathway 110 has a second end 110B that terminates at a heating vent 114 located on an inner wall 108A of the inner dome 108. The first airflow pathway 110 is configured to allow heated air from the heat source 206 to enter the cooking chamber 102. The second airflow pathway 112 has a second end 112B located at an upper portion of the upper assembly 100. As discussed in more detail below, the outer dome 106 includes a number of upper and lower (or exhaust) vents 118J, 118O to allow heated air to escape the oven 10. In one embodiment, as shown in FIG. 2A, the first and second airflow pathways 110, 112 are separated by an interior wall 111.

In operation, the ceramic cooking stone 72 of the second heat absorbing and emanating device 70 absorbs and retains heat from the heated air of the first airflow pathway 110.

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As shown in FIG. 1F, the thermometer 18 includes a dial 18A that includes an indicator (not shown), such as a radial indicator, to provide a visual indication of the temperature in the cooking chamber 102. The thermometer 18 further includes a stem 18B extending from the back of the dial 18A. The stem 18B is threaded at an upper end 18C located adjacent the dial 18A. A sensing element (not shown) is located within a lower end 18D of the stem 18B. A positioning pin 18E assists orientating the thermometer 18 within the upper assembly 100.

With reference to FIGS. 3A-3B, the outer dome 106 includes an outer dome top panel 118, an outer dome front panel 120, an outer dome first side panel 122, an outer dome second side panel 124 and an outer dome back panel 126. In one embodiment of the present invention, the outer dome panels 118, 120, 122, 124, 126 are composed from stainless steel and are connected using fasteners, such as screws, to form the outer dome 106.

With respect to FIGS. 3C, 3D and 3E, in the illustrated embodiment, the outer dome top panel 118 has a generally rectangular outer shape formed by first, second, third and fourth lower sides 118A, 118B, 118C, 118D. A rectangular top surface 118E forms the top of the outer dome top panel 118. Upper edges of each of the first, second, third and fourth sides 118A, 118B, 118C, 118D are connected to outer sides of the rectangular top surface 118E by respective first, second, third and fourth sloping side walls 118F, 118G, 118H, 118I.

One or more of the sloping side walls 118F, 118G, 118H, 118I may include a plurality of upper vents 118J. As shown in FIG. 3E, first, second, third and fourth flanges 118K, 118L, 118M, 118N extend inward from the respective first, second, third and fourth sides 118A, 118B, 118C, 118D. One or more of the flanges 118K, 118L, 118M, 118N may include a plurality of downwardly facing lower vents 118O. Returning to FIG. 2, the upper vents 118J and the lower vents 118O are located at the end of the second airflow pathway 112. The upper and lower vents 118J, 118O act as exhaust vents by providing a pathway for heated air to exit the oven 10.

With reference to FIGS. 3F and 3G, the outer dome front panel 120 includes an outer dome front inner side 120A. The outer dome front inner side 120A includes a door cutout 120B for receiving the first and second doors 12A, 12B. In the illustrated embodiment, a plurality of flanges 120C extend away from (inwardly or outwardly) the edges of the outer dome front inner side 120A. The flanges 120C may include one or more apertures for receiving fasteners (not shown) to connect the outer dome front panel 120 to the outer dome top panel 118, the outer dome first and second side panels 122, 124 and/or the lower assembly 200. The hinge 14 may be mounted to hinge support brackets 120D which are connected to the flange 120C extending inwardly from the door cutout 120B. An outer dome front panel support 120E connected to the flange 120C extending inwardly from the door cutout 120B may provide additional support to the first and second doors 12A, 12B and the hinges 14. The front inner panel 12D may further include a thermometer cutout 120F for receiving the thermometer 18.

With reference to FIG. 3H, the outer dome back panel 126 includes an outer dome back side 126A and a plurality of outer dome back panel flanges 126B extending away from the edge of the outer dome back side 126A. The outer dome back panel flanges 126B may include one or more apertures or slots 126C for receiving fasteners (not shown) to connect the outer dome back panel 126 to the outer dome side panels 122, 124 and/or the lower assembly 200. In the illustrated embodiment, a plurality of rivet nuts 126D may be coupled

to a lower one of the outer dome back panel flanges 126B for receiving appropriate fasteners (not shown).

With reference to FIG. 3I, a perspective view of the outer dome second side panel 124 is shown. The outer dome first side panel 122 is similar. The outer dome second side panel 124 includes an outer dome side panel side 124A and outer dome side panel extensions 124B. A plurality of flanges 124C extend from top and bottom edges of the outer dome side panel side 124A and from the outer dome side panel extensions 124B. The flanges 124C may include apertures 124D or tabs with apertures 124E for receiving fasteners (not shown) to connect the outer dome side panel 124 to the outer dome top panel 118, the first and second side panels 122, 124, and the lower assembly 200.

With reference to FIGS. 4A-4G, the first door 12A includes an outer door panel 12C and an inner door panel 12D. As shown, the hinge 14 includes an upper hinge portion 14A and a lower hinge portion 14B. The upper hinge portion 14A includes apertures 14C for receiving fasteners (not shown) to couple the first door 12A to an inner surface of the outer door panel 12C. The lower hinge portion 14B includes a plurality of apertures 14D for receiving fasteners (not shown) to couple the first door 12A to the outer dome front panel 120. With reference to FIGS. 4C and 4D, the inner door panel 12D includes a flange 12E that extends from the outer periphery of the inner door panel 12D towards the outer door panel 12C to form an inner door cavity 12F between the outer and inner door panels 12C, 12D. A notch 12M within the flange 12E receives the lower hinge portion 14B. A plurality of door brackets 12H (FIG. 4E) and a plurality of insulation panels 12I are located within the inner door cavity 12F. A seal 12J may be affixed to at least a portion of an outer surface of the flange 12E. In the illustrated embodiment, the seal 12J has a diameter of 0.5 inches. The insulation panels 12I and the seal 12J help to retain heat within the cooking chamber 102. The inner door panel 12D and the flange 12E include a plurality of apertures 12K for receiving respective fasteners to connect the door brackets 12H, the outer door panel 12C and the hinge 14 to the inner door panel 12D. The outer door panel 12C includes a door handle aperture 12L for receiving the handle 16.

The second door 12B is similar, but in an opposite configuration, to the first door 12A, and therefore not shown in greater detail.

As mentioned above, the first and second doors 12A, 12B may be removable. The upper hinge portion 14A allows the respective first or second door 12A, 12B to be opened and closed and is removable from the lower portion 12B by lifting the first or second door 12A, 12B and the upper hinge portion 14A then the first or second door 12A, 12B is in the open position.

As shown in FIGS. 1A-4B, each handle 16 includes a handle base 16A, an insulation washer 16B and a handle portion 16C. The handle base 16A includes a handle base first portion 16A-1 is fastened to the respective first or second door 12A, 12B and extends outward therefrom. The insulation washer 16B is positioned between the handle base first portion 16A-1 and the outer door panel 12C. The handle base 16A includes a handle base second portion 16A-2 that is coupled to the handle base first portion 16A-1 and extends at a right angle therefrom as shown. In the illustrated embodiment, the handle portion 16C is located at an opposite end of the handle base second portion 16A-2. The handle portion 16C may have a textured surface, such as having grooves or ridges, to provide a gripping surface.

With reference to FIGS. 5A-5G, the inner dome 108 includes an inner dome upper portion 128 and a lower dome

portion 130. A plurality of insulation pads 132 are affixed, for example, by adhesive, to an outer surface 128A of the inner dome upper portion 128 and the lower dome portion 130 to assist in keeping heat within the cooking chamber 102. A pair of halogen light assemblies 134 are mounted to the outside of the inner dome upper portion 128 which provide lighting to the cooking chamber 102. A plurality of insulation support brackets 136 mounted to the inner dome upper portion 128 by fasteners help maintain the insulation panels 132 mounted to the inner dome upper portion 128 in place.

With specific reference to FIGS. 5A-5D, the inner dome lower portion 130 includes an inner dome lower portion front panel 130A, inner dome lower portion back panel 130B and first and second inner dome lower portion side panels 130C, 130D. The inner dome lower portion front, back and side panels 130A, 130B, 130C, 130D are composed from stainless steel held together using appropriate fasteners (not shown). Flanges 130E extend from the inner dome lower portion side panels 130A, 130B, 130C, 130D include apertures for receiving appropriate fasteners to connect the inner dome lower portion 130 to the inner dome upper portion 128 and to the lower assembly 200. The inner dome lower portion front panel 130A includes a door cutout 130F that is aligned with the door cutout 120B in the outer dome front panel 120. The inner dome lower portion side panels 130C, 130D form part of the inner wall of the inner dome 108A (see above). In the illustrated each inner dome lower portion side panels 130C, 130D includes an exhaust vent 114. A first airflow pathway 110 is located on each side of the upper assembly 100 terminating at the respective heating vent 114.

With specific reference to FIGS. 5A, 5B, 5E and 5F, the inner dome upper portion 128 includes an upper portion top surface 128A. In the illustrated embodiment, the inner dome upper portion 128 has a generally rectangular shape define by the upper portion top surface 128A and an upper portion flange 128B. A plurality of upper portion side panels 128C extend from the upper portion top surface 128A to the upper portion flange 128B. The upper portion top surface 128A, upper portion flange 128B and upper portion side panels 128C are composed from stainless steel held together using appropriate fasteners (not shown).

The upper portion top surface 128A includes a cooking stone opening 128D. As shown in FIGS. 2A and 2B, the ceramic cooking stone 72 of the second heat absorbing and emanating device 70 rests on the inner dome upper portion 128 such that the ceramic cooking stone 72 is exposed to the cooking chamber 102. An inner dome upper portion 128 is fastened to the upper portion top surface 128A and retains the ceramic cooking stone 72 in place.

The pair of halogen light assemblies 134 are mounted to opposites upper portion side panels 128C. Light from the halogen light assemblies 134 enters the cooking chamber 102 through a pair of respective light openings 128E in the upper portion side panels 128C.

An inner dome heat exhaust panel 140 is fastened above the inner dome stone cover 138. Air from the cooking chamber 102 exhausts through an exhaust opening 128F and is routed towards the lower and upper vents 118O, 118J by an inner dome heat exhaust panel 140 fastened to the outer dome top panel 118 above the inner dome stone cover 138.

With reference to FIGS. 1D-1E, 6A-6B, 12 and 13, the oven 10 includes a control panel assembly 142 and a drawer assembly 144. The control panel assembly 142 provides controls to allow the burners 206A, 206B to be ignited and adjusted to heat the cooking chamber 102 to a desired

temperature (see below). The drawer assembly **144** includes a drawer **144A** in which food may be placed and kept warm (see below). In the illustrated embodiment the drawer **144A** may have a depth of 15" and a depth of 14". The temperature within the drawer assembly **144** is dependent upon the temperature within the cooking chamber **102**. In the illustrated embodiment, the ambient temperature within the drawer may range from 80-200 degrees Fahrenheit.

The control panel assembly **142** includes first and second knob assemblies **20**, **22** and first and second ignition buttons **24**, **26**. Each burner knob assembly **20**, **22** includes a respective rotatable knob **20A**, **22A** and bezel **20B**, **22B**. The first burner knob assembly and first and second ignition buttons **24**, **26** may be used to ignite the first or main gas burner **206A** and the second burner knob assembly and first and second ignition buttons **24**, **26** may be used to ignite the second or rear gas burner **206B**. For example, to ignite the first gas burner **206A**, the first burner knob assembly **20** must be rotated to an ignition position (as indicated on the respective bezel **20B**, **22B**) and pushed inward (to allow gas to flow to the main gas burner **206A**). Then the first ignition button **24** may be pressed to ignite the gas. Once the gas burner **206A**, **206B** has been ignited, the burner knob assemblies **20A**, **20B** to the desired level. The control panel assembly **142** further includes a see-through slide indicator **28**. The see-through slide indicator **28** includes an indicator knob **28A**. Indicator knob **28A** is moveable between an open position and a closed position (shown in FIG. 1A). When the indicator knob **28A** is in the open position, the interior of the lower assembly **200** is viewable such that visual configuration that the gas burner **206A** is lit may be seen through a viewing glass (not shown).

With reference to FIG. 13, the drawer assembly **144** includes the drawer **144A**, a drawer front panel **144B**, a drawer handle **144C** and a warming rack **144D**. The warming rack **144D** fits within, and is removable from, the drawer **144A**. The drawer front panel **144B** is mounted to the drawer **144A** and the drawer handle **144C** is mounted to the drawer front panel **144B**.

With reference to FIGS. 6A-6C, 7A-7D, 8A-8F and 9A-9E, the lower assembly **200** includes an outer enclosure **300**, an inner enclosure **400** and the fire box **204**. The lower assembly **200** is shown in FIG. 6C without the enclosure **116**, the drawer assembly **144**, and the ceramic cooking stones **62** to illuminate the interior of lower assembly **200**. With specific reference to FIG. 6C, a pair of stone main supports **208** are mounted to the fire box **204** to support the ceramic cooking stones **62**.

With specific reference to FIGS. 7A-7C, the outer enclosure **300** includes an outer enclosure back panel **302** and first and second outer enclosure side panels **304**, **306**. The outer enclosure back panel **302** and first and second outer enclosure side panels **304**, **306** are composed from stainless steel held together using appropriate fasteners (not shown). The first and second outer enclosure side panels **304**, **306** include a plurality of first and second outer enclosure side panel flanges **304A**, **306A** and a plurality of side panel vents **304B**, **306C**. The first and second outer enclosure side panel flanges **304A**, **306A** includes a plurality of apertures for receiving the fasteners. As explained in further detail below, the side panel vents **304B**, **306B** allow external ambient air to enter the internal cavity **202** of the lower assembly **200**.

The outer enclosure back panel **302** includes an opening **302A**. Power is provided to the oven **10** via a transformer **312**. The transformer **312** is mounted to a transformer cover **314** which is mounted to the outer enclosure back panel **302** adjacent the opening **302A**. An outer back cover plug cover

308 is also mounted to the outer enclosure back panel **302** to provide protection to the internal cavity **202**. A cable grommet **310** is positioned within an aperture in the outer back cover plug cover **308**. The cable grommet **310** allows passage of a transformer power cord **316** (only a portion of which is shown).

With specific reference to FIGS. 8A-8H, the inner enclosure **400** includes heat shield **402**, a fire box support structure **404** and a crumb collection assembly **406**. The heat shield **402** provides thermal isolation between the fire box **204** and the rest of the lower assembly **200**. The fire box **204** is supported by the fire box support structure **404**. The inner enclosure **400** is shown without the heat shield **402** in FIGS. 8B and 8C.

The crumb collection assembly **406** allows crumbs funneled from the top of the cooking stones **62** (via opening **62A** in one of the cooking stones **62**) into a crumb collection slot **406A** down a crumb collection chute **406B** and into a removable crumb collection drawer **406C**. The removable crumb collection drawer **406D** slides along a pair of crumb collection drawer rails **406D** mounted to a bottom surface of the fire box support structure **404** (see FIG. 8H). In the illustrated embodiment, the inner enclosure **400** includes a pair of spaced apart bottom support brackets **408** mounted across the bottom of the inner enclosure **400**. A pair of spaced apart storage drawer brackets **410** are mounted to the bottom support brackets **408**. A pair of storage drawer slides **412** are mounted to the bottom bracket **408**. The drawer assembly **144** is mounted or connected to the drawer slides **412** to allow the drawer assembly **144** to be moved from open and closed positions.

In the illustrated embodiment, the heat shield **402** includes an upper vertical portion **402A** and a lower horizontal portion **402B**. The upper vertical portion **402A** extends from a back edge of the lower horizontal portion **402B** at a right angle. With specific reference to FIGS. 8D-8F, the heat shield **402** includes a lower pan **402C**, an insulation panel **402D**, and an upper pan **402E**. The lower pan **402C** and the upper pan **402E** each have an inner cavity **202** and fit together to fully enclose the insulation panel **402D**. In one embodiment of the present invention, the lower pan **402C** and the upper pan **402E** composed of stainless steel and are fastened together using a plurality of fasteners (not shown). A heat shield front panel **402F** is mounted to the fire box support structure **404**.

As shown in FIGS. 8A-8B and 8G-8H, the fire box support structure **404** includes a number of fire box support brackets **414** mounted to an interior of the fire box support structure **404**. The fire box **204** rests within the fire box support structure **404** and is supported by the fire box support brackets **414**.

In the illustrated embodiment, the first box support structure **404** includes a fire box support structure top panel **416** and a pair of fire box support structure side panels **418**. In the illustrated embodiment, the fire box support structure top panel and side panels **416**, **418** are composed from stainless steel and fastened together using a plurality of fasteners (not shown).

As shown, in the illustrated embodiment the fire box support structure top panel **416** includes a plurality of vents **416A** located on opposite sides. Further, each fire box support structure side panel **418** includes a plurality of lower vents **418A** and a plurality of upper vents **418B**.

With reference to FIGS. 9A-9E, the illustrated embodiment the fire box **204** is composed from stainless steel and includes a fire box top panel **210**, a fire box rear panel **212**, a fire box front panel **214**, opposing fire box side panels **216**,

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and a bottom panel **218**. The fire box top panel **210** includes a plurality of fire box top panel vents **210A** and the fire box rear panel **212** includes a plurality of fire box rear panel vents **212A**. Additional, each of the side panels **216** includes a plurality of fire box side panel vents **216A**.

Air enters the oven **10** from below the oven **10** and/or through the vents **306B** in the side panels **316** of the outer enclosure **300** and the lower and upper vents **418A**, **418B** of the fire box support structure side panels **418**. Then, air enters the fire box **204** via vents **212A**. The first and second airflow pathways **110**, **112** start within the fire box **204** and exit the fire box **204** through the vents **216A** in the fire box side panels **216**. Air may also enter the oven **10** via gaps or vents located around the control panel **142A**.

The first and second pathways **110**, **112** proceeds through the vents **216A** at either side of the top panel **216** of the fire box **204**. As shown in FIGS. **2A-2B**, the first pathway **110** traverses between the outer dome **106** and the inner dome **108**. The first pathway **110** ends at the heating vents **114** in the inner walls **108A** of the inner dome **108**. The second pathway **112** also traverses between the outer dome **106** and the inner dome **108** and terminates at one of the upper and lower vents **118J**, **118O** in the outer dome top panel **118**.

As discussed previously, gas is provided to the oven **10** via the swiveling elbow connector **116D**. With reference to FIG. **10**, gas is controllably provided to the first and second gas burners **206A**, **206B** using the first and second burner knob assemblies **20**, **22** and a valve assembly **320**. The valve assembly **320** includes a gas manifold **322** and first and second manifold burner valves **324**, **326**. The gas manifold **322** is connected to the swiveling elbow connector **116D** via a first hose adapter **328**, a first corrugated hose **330** and a second hose adapter **332**. The first and second burner knob assemblies **20**, **22** are mounted to respective stems **324A**, **326A** of the first and second manifold burner valves **324**, **326**. The gas manifold **322** is mounted to the fire box **204** via a pair of gas mounting brackets **338**.

In the illustrated embodiment, the main gas burner **206A** is an H-shaped burner having rated at **18,000 BTU**. A burner air shutter **340** is connected to an input end of the H-shaped gas burner **206A**. A burner orifice connector **342** is connected to the burner air shutter **340**. A second corrugated hose (not shown) is between an output end **324B** of the first manifold burner valve **324** and the burner orifice connector **342** via a third hose connector **334**.

A first igniter assembly **344** is coupled to the first burner knob assembly **20**. The first burner knob assembly **20** controllably supplies gas to the first gas burner **206A** which then may be ignited by the first igniter assembly **344** using the first ignition button **24**.

With reference to FIGS. **10** and **11**, in the illustrated embodiment a second burner air shutter is connected to an end of the second burner **206B**. A second burner orifice connector **348** is connected to the second burner air shutter **346**. A third corrugated hose (not shown) is connected between an output end **326B** of the second manifold burner valve **326A** (via a fourth hose adapter **336**) and the second burner orifice connector **348**. A second igniter assembly **350** is coupled to the second burner knob assembly **22**. The second burner knob assembly **22** controllably supplies gas to the second gas burner **206B** which then may be ignited by the second igniter assembly **350** using the second ignition button **26**.

The invention has been described in an illustrative manner, and it is to be understood that the terminology which has been used is intended to be in the nature of words of description rather than of limitation. It is now apparent to

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those skilled in the art that many modifications and variations of the present invention are possible in light of the above teachings. It is, therefore, to be understood that the invention may be practiced otherwise than as specifically described.

What is claimed is:

1. An oven, comprising:

an upper assembly having an upper portion and a bottom, and the upper assembly forming a cooking chamber and having a front side with an open entry to the cooking chamber;

a lower assembly located below the upper assembly with the lower assembly having an internal cavity;

a fire box located within the internal cavity of the lower assembly and having a first heat source;

a heat absorbing and emanating device located between the upper and lower assemblies and being configured to absorb heat from the heat source and emanate heat into the cooking chamber of the upper assembly; and

a see-through slide indicator having an indicator knob mounted within a single opening in the lower assembly, the indicator knob moveable within the single opening between an open position and a closed position with the firebox viewable within the internal cavity through the single opening when the indicator knob is in the open position.

2. An oven, comprising:

an upper assembly having an upper portion and a bottom, and the upper assembly forming a cooking chamber and having a front side with an open entry to the cooking chamber;

a lower assembly located below the upper assembly with the lower assembly having an internal cavity;

a fire box located within the internal cavity of the lower assembly and having a first heat source; and,

a heat absorbing and emanating device located between the upper and lower assemblies and being configured to absorb heat from the heat source and emanate heat into the cooking chamber of the upper assembly,

wherein the upper assembly includes an outer dome and an inner dome with the inner dome being located within the outer dome and forming first and second airflow pathways disposed between the inner and outer domes, each of the first and second airflow pathways having an open end located at the bottom of the upper assembly and being open to the internal cavity of the lower assembly with the first airflow pathway having a second end terminating at a vent located on an inner wall of the inner dome and configured to allow heated air from the heat source to enter the cooking chamber, and the second airflow pathway having a second end terminating in the upper portion of the upper assembly,

wherein the inner dome of the upper assembly further includes an inner dome upper portion coupled to the inner wall of the inner dome, with the inner dome upper portion having a top surface and a plurality of side panels extending from the top surface, and

wherein the top surface is closed such that the heated air within the cooking chamber of the inner dome is unable to exit through the top surface of the inner dome upper portion.

3. An oven, as set forth in claim 2, wherein the heat absorbing and emanating device includes at least one ceramic stone.

4. An oven, as set forth in claim 2, further including a second heat source located within the upper assembly for heating the cooking chamber.

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5. An oven, as set forth in claim 4, further including a heat guard located between the second heat source and the cooking chamber.

6. An oven, as set forth in claim 2, wherein the upper assembly includes at least one exhaust vent located within the upper portion of the upper assembly near the second end of the second airflow path for allowing heated gas to exit the oven.

7. An oven, as set forth in claim 2, further including a thermometer for sensing a temperature associated with the cooking chamber.

8. An oven, as set forth in claim 7, wherein the thermometer includes an indicator for providing a visual indication of the temperature and a stem extending from the indicator into the cooking chamber.

9. An oven, as set forth in claim 2, further including at least one removable door rotatably coupled to the upper assembly within the open entry.

10. An oven, as set forth in claim 2, wherein the inner dome includes at least one insulation pad configured to retain heat within the cooking chamber.

11. An oven, as set forth in claim 2, wherein the outer dome forms an exterior of the upper assembly, the inner dome forms at least a portion of the cooking chamber, the first and second airflow pathways being located between the inner and outer domes.

12. An oven, as set forth in claim 11, wherein the inner dome includes at least one lower portion side panel, wherein the first and second airflow pathways are located between the exterior of the upper assembly and the at least one lower portion side panel.

13. An oven, as set forth in claim 11, wherein the inner dome includes first and second lower portion side panels located on opposite sides of the inner dome, wherein the first and second airflow pathways are located between the exterior of the upper assembly and each lower portion side panel.

14. An oven, as set forth in claim 13, wherein the heating vent is located within each of the side panels.

15. An oven, as set forth in claim 11, wherein the outer dome includes an outer dome top panel, the outer dome top panel having a rectangular shape defined by front, back and first and second side panels, the outer dome top panel further including a rectangular top surface and a plurality of sloping walls, each sloping wall extending from an edge of the rectangular top surface to a respective one of the front, back and side panels.

16. An oven, as set forth in claim 15, including a first exhaust vent located in one of the sloping walls, the second airflow path ending at the first exhaust vent.

17. An oven, as set forth in claim 16, the outer dome top panel including at least one flange facing downward and extending from a respective one of the front, back and side panels, the at least one flange including a second exhaust vent.

18. An oven, as set forth in claim 16, wherein one of the plurality of side panels includes an exhaust opening to permit heated air to exit the cooking chamber and route the heated air towards the first exhaust vent located in one of the sloping walls of the outer dome.

19. An oven, as set forth in claim 15, wherein the upper assembly further includes a second heat absorbing and emanating device within the second airflow pathway and located adjacent a top portion of the cooking chamber.

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20. An oven, as set forth in claim 2, wherein one of the plurality of side panels includes an exhaust opening to permit heated air to exit the cooking chamber.

21. An oven, as set forth in claim 2, wherein the top surface includes an opening, and further comprising a ceramic cooking stone resting upon the top surface such that the ceramic cooking stone is exposed to the cooking chamber.

22. An oven, comprising:

an upper assembly having an upper portion and a bottom, and the upper assembly forming a cooking chamber and having a front side with an open entry to the cooking chamber;

a lower assembly located below the upper assembly with the lower assembly having an internal cavity;

a fire box located within the internal cavity of the lower assembly and having a first heat source;

a second heat source located within the upper assembly for heating the cooking chamber;

a heat guard located between the second heat source and the cooking chamber;

a thermometer coupled to the lower assembly for sensing a temperature associated with the cooking chamber;

at least one removable door rotatably coupled to the upper assembly within the open entry; and,

a heat absorbing and emanating device located between the upper and lower assemblies and being configured to absorb heat from the heat source and emanate heat into the cooking chamber of the upper assembly,

wherein the upper assembly includes an outer dome and an inner dome with the inner dome being located within the outer dome and forming first and second airflow pathways disposed between the inner and outer domes, each of the first and second airflow pathways having an open end located at the bottom of the upper assembly and being open to the internal cavity of the lower assembly with the first airflow pathway having a second end terminating at a vent located on an inner wall of the inner dome and configured to allow heated air from the heat source to enter the cooking chamber, and the second airflow pathway having a second end terminating in the upper portion of the upper assembly, wherein the upper assembly further includes at least one exhaust vent located within the upper portion of the upper assembly in the outer dome near the second end of the second airflow path for allowing heated gas to exit the oven, wherein the outer dome forms an exterior of the upper assembly, the inner dome forms at least a portion of the cooking chamber, the first and second airflow pathways being located between the inner and outer domes,

wherein the inner dome of the upper assembly further includes an inner dome upper portion coupled to the inner wall of the inner dome, with the inner dome upper portion having a top surface and a plurality of side panels extending from the top surface, and

wherein the top surface is closed such that the heated air within the cooking chamber of the inner dome is unable to exit through the top surface of the inner dome upper portion.

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